# A Bayesian Method of Change-point Estimation with Recurrent Regimes: Application to GARCH Models

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**Abstract**

We present an estimation and forecasting method, based on a differential evolution MCMC method, for inference in GARCH models subjected to an unknown number of structural breaks at unknown dates. We treat break dates as parameters and determine the number of breaks by computing the marginal likelihoods of competing models. We allow for both recurrent and non-recurrent (change-point) regime specifications. We illustrate the estimation method through simulations and apply it to seven financial time series of daily returns. We find structural breaks in the volatility dynamics of all series and recurrent regimes in nearly all series. Finally, we carry out a forecasting exercise to evaluate the usefulness of structural break models.

*Keywords:* Bayesian Inference, MCMC, Structural Breaks, Recurrent Regimes, Marginal Likelihood, GARCH, Forecasting.

**JEL Classification:** C11, C15, C22, C53, C58.

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## 1. Introduction

The estimation of econometric models ignoring the presence of possible structural breaks, or more generally time-varying parameters, can result in several drawbacks. Importantly, it can produce poor forecasts as a result of the bias in the estimates and give the spurious impression of a nearly integrated behaviour of the data (see, e.g., Diebold, 1986; Lamoureux and Lastrapes, 1990; Hillebrand, 2005). In order to tackle these issues, more flexible models have been designed. They can be split into two categories: component and smooth transition models (e.g., Primiceri, 2005; Amado and Teräsvirta, 2008; Engle and Rangel, 2008) and abruptly switching models, namely Markov-switching models (e.g., Hamilton, 1989; Gray, 1996; Haas et al., 2004) and change-point models (e.g., Bai and Perron, 1998; Andreou and Ghysels, 2002; Pesaran et al., 2006). In this paper, we focus on change-point models but we allow for recurrent regimes, which makes the change-point specification more flexible than a Markov-switching scheme.

In Bayesian econometrics, dealing with breaks in Markov-switching or change-point models is usually done by treating as parameters the hidden state variables that identify the operating regime at each time point. The inference on the state variables is then performed by applying the forward-backward algorithm proposed by Chib (1996; 1998). This algorithm cannot be applied to models in which the likelihood function depends recursively on the past state variables. This issue, called the path dependence problem, makes the number of computations to be performed by the computer grow exponentially with the length of the time series. Moreover, subsequent improvements of Chib's algorithm by Koop and Potter (2007) and Giordani and Kohn (2008) do not solve the problem.

Our main contribution is to develop a Discrete-DREAM (D-DREAM hereafter) algorithm that enables us to treat directly the break dates as parameters, rather than to infer them as a by-product of the hidden Markov chain for the state variables. The algorithm, which is of the Metropolis type, is said to be "block-at-a-time" since we sample the parameters of the model (which constitute the first block) given the break dates, and the break dates (second block) given the parameters. Although both blocks could be sampled jointly using a Random Walk Metropolis (RWM) algorithm, we resort to this slightly more elaborate technique since it highly improves the MCMC mixing properties. The D-DREAM algorithm is a discretised version of the DREAM algorithm of Vrugt et al. (2009), which is limited to a continuous support.[1] The D-DREAM algorithm has the advantage to avoid the path dependence issue since break dates are treated as parameters and not as by-products of the hidden Markov chain of state variables.

Although our main goal is to estimate change-point models, we can also deal with recurrent regimes. More precisely, in this setup we can easily allow for the regimes to be active in several distinct periods of the sample. In short, it suffices to constrain the model parameters of non-subsequent regimes to be the same and to sample the two blocks as ex-

[1] The $DREAM_{(D)}$ algorithm of Vrugt and Ter Braak (2011) is an example of discretised version of the DREAM algorithm. In that respect, it should be noted that we developed our algorithm independently from Vrugt and Ter Braak (2011).



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plained above. Consequently, another contribution of this paper is to provide a framework for the analysis of processes halfway between change-point and Markov-switching specifications. The marginal likelihood criterion can be used to decide on the unknown number of breaks as well as on whether recurrent regimes are present in the sample, or whether the data are better described by a pure change-point model.

We apply the D-DREAM algorithm to GARCH models in order to show that it can deal with structural breaks in path dependent processes. We contribute to a recent literature. He and Maheu (2010) use a Sequential Monte Carlo *on-line* method, which is fast and better adapted to forecasting than in-sample detection of breaks. On the contrary, Liao (2008), Bauwens et al. (2010), and Bauwens et al. (2013) develop *off-line* MCMC methods. Similarly to the present work, Liao (2008) considers the break dates as parameters. She builds on Stephens (1994) to handle GARCH models and use sequential strategies to determine the number of breaks, which is different from our strategy of detecting breaks simultaneously. Still considering MCMC methods, the single-move MCMC algorithm for Markov-switching GARCH models in Bauwens et al. (2010) has the drawback to create strong dependence in the generated draws. This problem is circumvented in Bauwens et al. (2013) by embedding a Sequential Monte Carlo technique inside an MCMC algorithm, so that all state variables are generated as a single block. Although this approach works well for Markov-switching and change-point models and allows to compute marginal likelihoods for model selection, it is computationally much more demanding than the D-DREAM algorithm presented in this paper.

The paper is organized as follows. In section 2, we present the D-DREAM algorithm in the context of change-point GJR-GARCH models. In section 3, we show how to deal with recurrent regimes. In section 4, we explain how to determine the unknown number of breaks and unknown ordering of regimes by computing Bayes factors. Simulations illustrate the efficiency of the algorithm in section 5. In section 6, we apply the algorithm to daily data for the S&P 500 index and other financial time series. Section 7 shows that the detection of break dates and recurrent regimes can be useful in forecasting exercises, though further research is needed on this topic. Section 8 contains conclusions.

## 2. D-DREAM Algorithm

We denote by $Y_T = (y_1, ..., y_T)'$ a time series of $T$ observations. Instead of considering a $T \times 1$ vector of states as in Chib (1996), we focus on a smaller vector of break dates $\Gamma = (\tau_0, \tau_1, \tau_2, ..., \tau_K, \tau_{K+1})'$, where $\tau_0 = 0$, $\tau_{k-1} < \tau_k$, $\tau_{K+1} = T$, and the number of break dates $K$ is considered known for the moment. For instance, the break date $\tau_2$ indicates the end of regime 2. The break date parameters hence constitute a discrete vector. Figure 1 illustrates how the break dates split a time series of length $T$ into 3 regimes when $K = 2$.

For simplicity, we present the D-DREAM algorithm by considering a change-point GJR-GARCH(1,1), or CP-GJR-GARCH(1,1), specification though the algorithm can easily be adapted to other models or variants of the model presented below (see Section 6 for two examples). A CP-GJR-GARCH(1,1) model is characterized by the following set of



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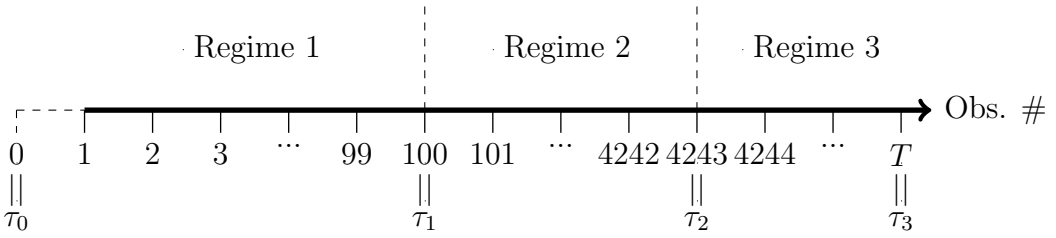



Figure 1: Example of a time series with 3 regimes.

equations, for all $t \in [1, T]$:

$$\begin{aligned} y_t &= \mu_k + \epsilon_t, \\ \epsilon_t &= \sigma_t \eta_t, \ with \ \eta_t \sim i.i.d. \, N(0,1), \\ \sigma_t^2 &= \omega_k + \alpha_k \epsilon_{t-1}^2 + \gamma_k \epsilon_{t-1}^2 \mathbb{1}_{\epsilon_{t-1}<0} + \beta_k \sigma_{t-1}^2, \end{aligned}$$

if $t \in ]\tau_{k-1}, \tau_k]$, with $k = \{1, 2, ..., K+1\}$ denoting the regime. We constrain some parameter spaces to accelerate the sampling from the posterior distributions but we allow for large supports so as not to bias results. More specifically, we assume that, $\forall k$, $\mu_k \in \Re$, $\omega_k \in [0, 10]$, $\alpha_k$ and $\gamma_k \in [-0.2, 1]$, $\beta_k \in [0, 1.2]$. We do not impose any stationarity condition and we impose the positivity of the variances. We impose that all conditional variances are positive by rejecting draws that imply negative conditional variances anywhere in the sample. We chose not to impose restrictions on the GJR-GARCH parameters themselves as it is important to allow for negative $\alpha$ parameters in the applications of section 6. For the S&P 500 series, the series for which the issue is most severe, we had to reject 0.36% of the draws. We detail our priors in subsection 2.4.

Let $\Theta = (\theta_1, \theta_2, ..., \theta_{K+1})'$, where $\theta_k = (\mu_k, \omega_k, \alpha_k, \gamma_k, \beta_k)$ contains the parameters of regime $k$, denote the set of parameters characterising the CP-GJR-GARCH(1,1) model. Our MCMC scheme simply consists in drawing iteratively from the conditional posterior of the model parameters $\Theta$, $\pi(\Theta|Y_T, \Gamma)$, and from the conditional posterior of the break dates $\Gamma = (\tau_1, \tau_2, ..., \tau_K)'$, $\pi(\Gamma|Y_T, \Theta)$.

### *2.1. Sampling the Model Parameters* $\Theta$

Since the full conditional posterior distribution of $\Theta$ has no standard parametric form, we resort to a Metropolis-Hastings algorithm to obtain draws from it. However, in a Metropolis-Hastings algorithm, convergence towards the limiting distribution can be very slow if the proposal distribution is not appropriate. Therefore, Vrugt et al. (2009) propose to run multiple chains in parallel and let these chains interact to propose new values for the parameters, without resorting to a fixed proposal distribution. They illustrate that their Differential Evolution Adaptive Metropolis (DREAM) algorithm significantly improves the efficiency of the MCMC simulations, compared with a random walk Metropolis algorithm and other adaptive schemes, by automatically tuning the scale and orientation of the



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proposal distribution. They show in particular that the DREAM algorithm is especially useful for high-dimensional and multi-modal problems.

Following the DREAM algorithm, the proposal of a new location for a given chain is constructed from the current locations of other chains. Focusing on the $i^{th}$ chain ($i = 1, 2, ..., N$) at the $j + 1^{th}$ MCMC iteration, the procedure can be described in 2 steps:

1. Propose a new $\Theta_i^{j+1}$, denoted by $Z_i$ hereafter, according to the following rule (omitting superscripts referring to iterations for simplicity):

$$Z_i = \Theta_i + \gamma(\delta, d)(\sum_{g=1}^{\delta} \Theta_{r_1(g)} - \sum_{h=1}^{\delta} \Theta_{r_2(h)}) + \zeta, \tag{1}$$

   with $g, h = 1, 2, ..., \delta$, $i \neq r_1(g)$, $r_2(h)$; $r_1(.)$ and $r_2(.)$ stand for random integers uniformly distributed on the support $[1, \ldots, i-1, i+1, \ldots, N]$ and it is required that $r_1(g) \neq r_2(h)$ when $g = h$; $\zeta \sim N(0, \eta_\Theta^2 I)$; and $d$ is the number of elements in $\Theta_i$ (and is equal to $5(K+1)$ in the case of GJR-GARCH(1,1) models). Notice that if $\gamma(\delta, d) = 0$ the algorithm boils down to a RWM scheme. However, since we run multiple chains in parallel, we take advantage of the locations of other chains by introducing the term $\gamma(\delta, d)(\sum_{g=1}^{\delta} \Theta_{r_1(g)} - \sum_{h=1}^{\delta} \Theta_{r_2(h)})$ and $\zeta$ becomes a small perturbation term present to ensure the convergence of the algorithm. Consequently, $\eta_\Theta^2$ is a small value compared with the width of the target distribution and we set $\gamma(\delta, d) = 2.38/\sqrt{2\delta d}$.[2] Note that the fixed number of pairs of chains, $\delta$, must be smaller than or equal to $(N-1)/2$.
2. Accept the proposed $Z_i$ according to the acceptance probability ratio of the Metropolis algorithm $\alpha(\Theta_i, Z_i) = \min\{\frac{f(Z_i|\Gamma_i, Y_T)}{f(\Theta_i|\Gamma_i, Y_T)}, 1\} = \min\{\frac{f(Y_T|Z_i, \Gamma_i)p(Z_i|\Gamma_i)}{f(Y_T|\Theta_i, \Gamma_i)p(\Theta_i|\Gamma_i)}, 1\}$.

It is important to realize that, in large dimensional spaces, it might not be optimal to update the entire vector of parameters as a whole. Therefore, we choose in the simulations and applications below to randomly break the large vector $\Theta_i$ into sub-vectors of smaller dimensions at each MCMC iteration and to apply the two steps above to each sub-vector in turn with the appropriate conditioning parameters. As initially proposed by Chib and Ramamurthy (2010), randomly choosing the size of the block that will be updated has several advantages over a one-block strategy such as avoiding some circumstances where the MCMC algorithm can be stuck and allowing for more efficient exploration of the posterior support. However these improvements come at a price since for each block, a Metropolis ratio must be computed. It therefore increases the computational time. As a last remark, notice that since the support of proposal (1) is not constrained, we map the constrained space of the model parameters into the real space: if $x$ is a parameter in $]x_{min}, x_{max}[$, we transform it to $\log[(x - x_{min})/(x_{max} - x)]$ in $\Re$.

[2] We follow Roberts and Rosenthal (2001) and Vrugt et al. (2009) for the choice of $\gamma(\delta, d)$. Note that if $\delta = 1$, then $\gamma(1, d) = 2.38/\sqrt{2d}$ is optimal for Normal posteriors as regards acceptance rates for $N \to \infty$, as shown in Ter Braak (2006).



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*2.2. Sampling the Break Dates* $\Gamma$

The sampling procedure of the break date vector $\Gamma$ is similar to the sampling of $\Theta$ and can be described in the same two steps. The only difference is that we need to round the elements of the proposed vector of break dates $Z_i$ so as to have integers. Proposal (1) then becomes:

$$Z_i = \Gamma_i + round[\gamma(\delta, K)(\sum_{g=1}^{\delta} \Gamma_{r_1(g)} - \sum_{h=1}^{\delta} \Gamma_{r_2(h)}) + \xi], \tag{2}$$

with $round[.]$ meaning that we take the nearest integer and $\xi \sim N(0, \eta_\Gamma^2 I)$. Note that, conditional on $K+1$ regimes, we draw $K$ break date(s) and $K+1$ set(s) of model parameters, which ensures that all regimes are forced to be visited.

Proposition 1 ensures the convergence and ergodicity of the chains in the D-DREAM algorithm.

**Proposition 1.** Consider a change-point model with a given number of breaks $K$. Then, $\forall i = 1, ..., N$, let $X_i = (\Theta_i', \Gamma_i')'$. The D-DREAM algorithm yields a Markov chain that is ergodic with unique stationary distribution, such that

$$(X_1, ..., X_N) \sim \tilde{\pi}(X_1, ..., X_N | Y_T) = \pi(X_1|Y_T) \times ... \times \pi(X_N|Y_T).$$

**Proof.** See Appendix A.

Intuitively, the D-DREAM algorithm works because it does not behave very differently from a simple Random Walk (RW) Metropolis algorithm. Note that a simple Metropolis ratio is needed to accept or reject the $\Theta$ or $\Gamma$ proposals as the new parameters for a given chain are generated from a symmetric proposal distribution. In proposition 1, we show that the detailed balance of the D-DREAM algorithm is ensured with a Metropolis ratio since the $\zeta$ and $\xi$ terms, respectively from (1) and (2), are drawn from Normal distributions and because the pairs of chains used to generate the proposals are randomly selected. In particular, if only one pair of chains is chosen ($\delta = 1$), then it could happen with equal probability that chain $a$ is selected and then chain $b$, so that the proposal would be a function of the difference between the locations of chains $a$ and $b$ ($a$-$b$), or that chain $b$ is selected before chain $a$ and that the proposal would be a function of the difference $b$-$a$, implying that the proposal can go in both directions. Following from the random selection of chains and their order to generate proposals, we also show in proposition 1 that the joint stationary distribution of the Markov chains factorizes into $N$ (the number of chains) independent marginal posterior distributions of interest. The latter are however identical as all chains converge together to the same stationary distribution.

To show further the resemblance of the DREAM sampling procedure with a RW Metropolis algorithm, let us focus on the continuous parameter $\Theta$ and assume that the first two posterior moments exist and are denoted by $\bar{\mu} = E(\Theta|Y_T)$ and $\bar{\Sigma} = \text{Cov}(\Theta|Y_T)$. According to (1), once all chains have converged to the posterior distribution, we have that the current parameter of any independent chain is on average $\bar{\mu}$ and that the difference



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$(\sum_{g=1}^{\delta} \Theta_{r_1(g)} - \sum_{h=1}^{\delta} \Theta_{r_2(h)})$ is on average null with variance $2\delta\bar{\Sigma}$. Since the variance parameter $\eta_{\Theta}^2$ is small, proposal (1) looks like $Z_i = \bar{\mu} + \gamma(\delta, K)\eta$ with $\eta \sim N(0, 2\delta\bar{\Sigma})$. The DREAM algorithm thus behaves like a standard RW algorithm and related theory can be used to optimally set the user-defined value $\gamma(\delta, K)$. Following Roberts and Rosenthal (2001), the parameter $\gamma(\delta, K)$ is set to $2.38/\sqrt{2\delta d}$.

*2.3. Advantages of the D-DREAM Algorithm*

In addition to allowing us to use directly the break dates as parameters (thus avoiding the path dependence issue) and also recurrent regimes (see section 3), the D-DREAM sampling scheme has some other practical advantages compared with existing methods. Firstly, the D-DREAM algorithm is relatively simple due to the fact that we consider break dates directly as parameters and not a more complex structure entailing a hidden Markov chain. Consequently, the D-DREAM algorithm is a simple Metropolis sampling procedure that does not require the use of filtering methods such as the forward-backward scheme as in Chib (1996) or the PMCMC algorithm of Bauwens et al. (2013).

Secondly, the computational time needed to simulate posterior draws with the D-DREAM algorithm is significantly reduced in comparison with other methods cited above. Although Bauwens et al. (2013) show that their PMCMC scheme is as fast as the MCMC method of Bauwens et al. (2010) when considering effective posterior draws, their algorithm consists of sequential Monte Carlo simulations run at every iteration of an MCMC scheme. For example, for the S&P 500 series in section 6.1, the D-DREAM algorithm is 30 times faster than the PMCMC algorithm of Bauwens et al. (2013).

Thirdly, our framework is more flexible than existing models as any distribution can be assumed for the break date parameters. For instance, a geometric distribution could be chosen as in Chib (1996), or a Poisson distribution as in Koop and Potter (2007). Moreover, our framework also allows for hierarchical prior distributions for the GARCH parameters. One could specify, for instance, a common distribution for all regimes as in Pesaran et al. (2006) or an evolving distribution depending on the parameters of the previous regime as in Koop and Potter (2007).

Fourthly, the D-DREAM algorithm allows us to estimate all break points simultaneously, as opposed to a "one-at-a-time" (or sequential) estimation strategy. Stephens (1994), who extends the work of Carlin et al. (1992) to multiple change points, considers such a sequential strategy. The disadvantage of sequential strategies is that the locations of the breaks might not coincide in practice with the locations of the breaks when estimated simultaneously; for further discussions from a frequentist point of view, see also Bai and Perron 1998; 2003. Besides, as Chib (1998) (p.224) puts it:

> The first [weakness of this approach] arises from the fact that the change points [...] are simulated one at a time from the [...] full conditional distributions [...] rather than from the joint distribution [...]. Sampling the latter distribution directly, say through a Metropolis-Hastings step, is not practical because of the difficulty of developing appropriate proposal generators for the change points. Liu et al. (1994) in theoretical work and Carter and Kohn



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(1994), Chib and Greenberg (1995a) and Shephard (1994) in empirical work have shown that the mixing properties of the MCMC output is considerably improved when highly correlated components are grouped together and simulated as one block. Conversely, MCMC algorithms that do not exploit such groupings tend to be slow to converge.

In that respect, the D-DREAM algorithm samples the break dates as one block and a symmetric generic proposal distribution is automatically generated by the interaction of the parallel chains.

*2.4. Priors, Starting Points, and Burn-in*

We choose rather standard and weakly informative priors for the model parameters, as shown in Table 1. The prior for the $\mu$ parameters, the mean of the series in the different regimes, is a Normal distribution centered at 0 and with variance 10 and the priors for the GARCH parameters are uniform distributions.[3] Regarding the break dates, we choose a prior that complies with the modelling constraints. More precisely, the chronological order of the break dates needs always to be respected and, for any $k \in \{1, 2, ..., K\}$, no break date can lie outside the interval $[1, T-1]$. We work with the regimes durations, defined $\forall k \in [1, K+1]$ as $d_k = \tau_k - \tau_{k-1}$, and choose independent negative binomial prior distributions, as in Koop and Potter (2007). It would be equivalent to work with priors on break dates since the Jacobian of the transformation from durations to break dates is equal to one. Note that our priors for the regime durations tend to penalize (very) short (and very long) regimes since we use a negative binomial distribution, with a large variance though. With this type of priors, outliers do not result into spurious regimes and, hence, are not an issue in the applications of section 6. Priors for the model parameters $\Theta$ also imply that it is costly to introduce a new regime as the marginal likelihood of the model will be affected (see section 4 and subsection 6.4 for a discussion).

The starting point of an MCMC algorithm is another relevant practical issue since choosing a highly likely point with respect to the posterior distribution ensures faster convergence. Typically, MCMC algorithms start at the maximum likelihood estimates (MLE) but they cannot be readily computed in the case of structural breaks; see Augustyniak (2014). Therefore, in order to find good starting values, we randomly generate many break date vectors $\Gamma$ and estimate the associated MLE of the GJR-GARCH parameters. The initial points of the $N$ chains are the $N$ randomly generated vectors of break dates and associated parameters exhibiting the highest likelihoods. We also ensure that the break dates used as starting values by the chains are not equal so that the algorithm mixes well from the beginning.

We determine the length of the burn-in period using the criterion in Gelman and Rubin (1992) ($\hat{R} < 1.2$ for all parameters). At each iteration of the burn-in phase we replace

[3]Since the D-DREAM algorithm takes draws from the posteriors of the parameters mapped into the real line $\Re$, we need to take into account the Jacobian of the mapping function when evaluating the prior density at a given point.



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| CP-GJR-GARCH(1,1) Parameters | | |
|---|---|---|
| $\forall k \in [1, K+1]$ | $\mu_k \sim N(0, 10)$ | $\omega_k \sim Uniform[0, 10]$ |
| | $\alpha_k$ and $\gamma_k \sim Uniform[-0.2, 1]$ | $\beta_k \sim Uniform[0, 1.2]$ |
| Duration Parameters | | |
| $\forall k \in [1, K]$ | $d_k = \tau_k - \tau_{k-1} \sim NB_{tr}(\frac{5(T-\tau_{k-1})}{2(T-\tau_{k-1}-5)}, 1 - \frac{5}{T-\tau_{k-1}})\ \mathbb{1}_{[1,T-5-\tau_{k-1}-(K+1)+k]}$ | |
| | $d_{K+1} = T - \tau_K$ | |

Table 1: Prior Distributions. $NB_{tr}$ denotes the truncated negative binomial distribution. The parameters $r$ and $p$ are the usual parameters of the negative binomial distributions (not truncated) such that, $\forall k \in [1, K]$, $E[d_k] = \frac{T-\tau_{k-1}}{2}$ and $Var[d_k] = \frac{(T-\tau_{k-1})^2}{10}$. Priors for the regime durations imply that the chronological order of the break dates is always respected and that no break date lie outside the interval [1,T-5[, which also ensures the positivity of the negative binomial parameters, otherwise the prior is zero and the draw of the break date vector $\Gamma$ is rejected. If a drawn break date is not the final one it will leave enough space for the other break dates before the end of the sample.

outlier chains by the chain with the highest likelihood (plus a small random term) to ease convergence. Outlier chains are defined similarly as in Vrugt et al. (2009) as the ones with a data log-likelihood smaller than the third quartile minus 2 times the inter-quartile range.

## 3. Dealing with Recurrent Regimes

The D-DREAM algorithm as described in the previous section can only deal with change-point specifications. Hence, a natural extension is to allow for recurrent regimes, like in a Markov-switching framework. Therefore, we consider a permutation variable $\lambda$ which is a $(K+1) \times 1$ vector indicating which regime is active between two break dates. More precisely, at each MCMC iteration all possible combinations in $\lambda$ are considered and the algorithm randomly draws one combination (section 4 provides the drawing procedure). Sampling directly the parameter $\lambda$ is much faster than running the algorithm considering all possible combinations one by one and then computing marginal likelihoods. As an example, assuming that the number of regimes is four, all possible $\lambda$, generally denoted by $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$, are as follows: (1,2,3,4), (1,2,1,4), (1,2,3,1), (1,2,3,2), (1,2,1,2). The first possibility (1,2,3,4) is a pure change-point specification whereas the others are recurrent regime specifications. For instance, in the case of the combination $(1, 2, 1, 4)$ the first regime is said to be recurrent as it is active before the first break date and between the second and third break dates. Any other combination than listed above would imply label switching problems and are therefore eliminated. For example, the combination $(1, 3, 1, 3)$ is by no way different from labeling the regimes $(1, 2, 1, 2)$.

The introduction of the $\lambda$ parameter somewhat complicates the sampling of the model parameters $\Theta$. Imagine that at a given iteration $j$ we draw a combination $\lambda^j$ from its full conditional density $\pi(\lambda^j | Y_T, \Theta^{j-1}, \Gamma^{j-1}) \propto f(Y_T | \Theta^{j-1}, \Gamma^{j-1}, \lambda^j) p(\Theta^{j-1}, \Gamma^{j-1} | \lambda^j) p(\lambda^j)$ (where $p(\lambda^j)$ is the prior density) and propose a whole set of parameters $\Theta^j$ and break



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dates $\Gamma^j$ as described in section 2, that is, without yet taking into account that there might be recurrent regimes. If $\lambda^j$ implies, for instance, that the third regime is actually the same as the first, then the data likelihood does not depend on the parameters $\theta_3^j$ and the proposal $(\Theta^j)$ could be accepted while the model parameters of the inactive third regime are unrealistic. Therefore, following this strategy, the sampling of the set of model parameters of likely inactive regimes will be highly inefficient, and Bayes factors will put too much weight on recurrent regime with respect to pure change-point specifications if the true specification includes recurrent regimes.

We explain in subsection 4.1 below how we deal with the issue of drawing the model parameters conditional on the permutation variable $\lambda$. By introducing $\lambda$, and by showing in the next section how to accurately obtain draws from its posterior, we provide a formulation that encompasses both change-point and Markov-switching models.

## 4. Model Selection

From Sections 2 and 3, assuming that the number of lags in the CP-GJR-GARCH model is taken as given, the algorithm still requires the user to determine the number of breaks, $K$, and the combination of regimes, $\lambda$, which are typically unknown. In a purely Bayesian approach, we need to calculate the probabilities of the CP-GJR-GARCH models for all values of $K$ and $\lambda$. In this section, we first consider the case of a given number of breaks $K$, and we show how to compute Bayes factors of the models differing by the ordering of the regimes by sampling directly the permutation parameter $\lambda$. The sampling technique used is based on Carlin and Chib (1995). In a second step, we show how to compute the marginal likelihoods of the pure change-point models. Eventually, we are able to recover the marginal likelihoods of all models differing by the number of breaks or combinations of regimes.

### *4.1. Bayes Factors to Determine the Recurrence of Regimes*

Let us first assume that the number of breaks is given, so that we can focus our attention on the combination of regimes. Then, we only need to sample the permutation parameter $\lambda$. Recall, however, that inactive regimes imply a dimensional problem in the sampling of the model parameters $\Theta$ (models with more recurrent regimes need less parameters than models with less recurrent regimes or no recurrent regime at all). Nevertheless, Carlin and Chib (1995) suggest estimating Bayes factors by including all models of interest in an MCMC sampling scheme and by allowing the Markov chains to jump from model to model. In fact, Carlin and Chib's method simply requires the introduction of pseudo-prior distributions needed in our context since the prior of $\Theta$ depends on the permutation parameter $\lambda$, as explained below.

Strictly following Carlin and Chib's sampling scheme, pseudo-prior distributions should be thought of as linking densities between parameters associated with a given value $\lambda$ and another value $\tilde{\lambda} \neq \lambda$. That is, when a chain selects a combination $\lambda$, the parameters of the regimes associated with the combination $\tilde{\lambda}$ should be drawn from their pseudo-priors.



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Formally, we have the following full conditional distributions for $\Theta^{K,j}$ (the set of parameters associated with the number of breaks $K$ and the combination $\lambda^{K,j}$):

$$\pi(\Theta^{K,j}|\lambda,\Gamma^K,Y_T) \propto \begin{cases} f(Y_T|\Theta^{K,j},\Gamma^K,\lambda)p(\Theta^{K,j}|\lambda)p(\Gamma^K)p(\lambda) & \lambda=\lambda^{K,j}, \\ p(\Theta^{K,j}|\lambda), & \lambda\neq\lambda^{K,j}, \end{cases}$$

where $p(\lambda)$ is the prior of $\lambda$, and $p(\Theta^{K,j}|\lambda\neq\lambda^{K,j})$ is the pseudo-prior (or linking density).

However, the recurrent regimes offer the particularity that the same model parameters are present in the various $\Theta^{K,j}$. Hence, instead of drawing from the pseudo-priors the entire set $\Theta$ of model parameters associated with a "not-drawn" $\tilde{\lambda}$, we take the advantage of this particularity and plug directly the accepted draw of active regimes in all $\Theta^{K,j}$. Then, we draw from the pseudo-priors only the subsets of parameters $\theta$ that correspond to inactive regimes and plug them again in the other $\Theta^{K,j}$. We set pseudo-prior distributions as independent Normal distributions with the posterior mean and variance of the model parameters, which are estimated at the end of a (second) burn-in phase from the posterior draws of the pure change-point model, and we choose uniform prior weights on the permutation parameters $\lambda$ associated with recurrent regimes.

Finally, the draw of $\lambda^K$ is done according to the following probabilities: $\forall j=\{1,...,J\}$,

$$\pi(\lambda^K=\lambda^{K,j}|\Theta^K,\Gamma^K,Y_T)=\frac{f(Y_T|\Theta^K,\Gamma^K,\lambda^{K,j})p(\Theta^K|\lambda^{K,j})p(\lambda^{K,j})}{\sum_{s=1}^{J}f(Y_T|\Theta^K,\Gamma^K,\lambda^{K,s})p(\Theta^K|\lambda^{K,s})p(\lambda^{K,s})}, \tag{3}$$

where $\Theta^K$ is the full set of model parameters associated with the number of breaks $K$, and $p(\Theta^K|\lambda^{K,j})$ is the joint prior density of the model parameters evaluated at $\Theta^{K,j}$ multiplied by the joint pseudo-prior density of the model parameters of non-active regimes in $\lambda^{K,j}$.

To sum up, at a given iteration and for a given chain, the algorithm starts by drawing the combination parameter $\lambda$ according to (3). Then, a proposition is made for the model parameters of the active regimes according to (1) and the parameters of inactive regimes are drawn from their pseudo-priors. When all $G_1$ iterations are run, the posterior probability of the specific permutation $\lambda^{K,j}$ is estimated by $\sum_{g_1=1}^{G_1}\mathbb{1}_{\lambda_{g_1}=\lambda^{K,j}}/G_1\approx\pi(\lambda^{K,j}|Y_T)$. From the next subsection, it will be clear that our strategy to compute the marginal likelihood of all models builds upon reliable estimates of the Bayes factors of the pure change-point models. For a given $K$, in order to obtain a reliable estimate of $\pi(\lambda^{CP}|Y_T)$, the posterior probability of the pure change-point specification, we need to ensure that the pure change-point model is visited by the MCMC algorithm. This can be done by determining an accurate prior probability $p(\lambda^{CP})$ during the second burn-in phase. A simple and effective way to do that is by comparing the data likelihood implied by the different models (pure change-point and recurrent regime specifications) and offset the differences by playing on $p(\lambda^{CP})$.

### *4.2. Bayes Factors to Determine the Number of Breaks*

Given that we can compute Bayes factors of models with different regime combinations, we need to find a way to compute the marginal likelihood (or Marginal Log-Likelihood,



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MLL) of the pure change-point models to compare the posterior probabilities of models differing by their number of regimes and regime combinations.

We use the Bridge sampling technique of Meng and Wong (1996) (also described in Fruhwirth-Schnatter, 2004). Using the integral symbol to integrate out $\Gamma$ (like in the continuous case, for notational simplicity), the marginal likelihood can be written as $f(Y_T) = \int\int f(Y_T|\Theta,\Gamma)p(\Theta,\Gamma)d\Theta d\Gamma$. Taking a function $t(\Theta,\Gamma)$ and a proposal distribution $q(\Theta,\Gamma)$, we can define the following two quantities:

$$\begin{aligned} A &= \frac{1}{G_1}\sum_{g_1=1}^{G_1} t(\Theta^{g_1},\Gamma^{g_1})q(\Theta^{g_1},\Gamma^{g_1}), \\ A_1 &= \frac{1}{G_2}\sum_{g_2=1}^{G_2} t(\Theta^{g_2},\Gamma^{g_2})f(Y_T|\Theta^{g_2},\Gamma^{g_2})p(\Theta^{g_2},\Gamma^{g_2}), \end{aligned}$$

where $G_1$ and $G_2$ are large. The sequence $\{\Theta^{g_1},\Gamma^{g_1}\}_{g_1=1}^{G_1}$ is sampled from $\pi(\Theta,\Gamma|Y_T)$, which is done by the MCMC algorithm, and $\{\Theta^{g_2},\Gamma^{g_2}\}_{g_2=1}^{G_2}$ is sampled from $q(\Theta,\Gamma)$. Meng and Wong (1996) show that the marginal likelihood can be computed using the ratio of these two quantities as $A_1/A \approx_a f(Y_T)$. We use the asymptotically optimal choice of $t(\Theta,\Gamma)$ derived in Meng and Wong (1996) that minimizes the expected relative error of the estimator for i.i.d. draws from $f(\Theta,\Gamma|Y_T)$ and $q(\Theta,\Gamma)$: $t(\Theta,\Gamma) = (\pi(\Theta,\Gamma|Y_T) + q(\Theta,\Gamma))^{-1}$. The proposal distribution $q(\Theta,\Gamma)$ is given in Appendix B.

We also consider an importance sampling approach, which simply consists in setting $t(\Theta,\Gamma) = q(\Theta,\Gamma)^{-1}$. Since our choice of $t(\Theta,\Gamma)$ for the Bridge sampling estimator depends on $f(Y_T)$ through $\pi(\Theta,\Gamma|Y_T)$, the estimator needs to be initialised, which we do with the importance sampling estimator. We let the Bridge sampler run for 50 iterations, each time using the marginal likelihood estimation of the previous iteration.

## 5. Simulations

In this section, we resort to Monte Carlo simulations to show that the D-DREAM algorithm performs well in practice. Two sets of experiments are performed. The first one shows that the algorithm is able to detect parameter changes in the volatility dynamics, even if the changes are small. The second one shows that the algorithm can deal with recurrent regimes.

### *5.1. Monte Carlo Simulations 1: Changes in Parameters*

We simulate 1000 data points with 32 different data generating processes (DGP) of CP-GJR-GARCH(1,1) models, most of which entail a break in the middle of the sample (500th observation). The 16 first DGPs deal with breaks in the $\omega$ parameter of the volatility equation. The aim is to show that the D-DREAM algorithm can detect changes in the level of the unconditional variance. The 16 other DGPs deal with breaks in the $\beta$ parameter. The goal is to show that the algorithm can also detect changes in the persistence of the



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volatility. We simulate 100 time series for each DGP and after each simulation we estimate the marginal likelihood of two models: a model without structural break and a model with one change point.[4] We draw 1000 posterior values of the parameters per chain after convergence, and the number of chains is equal to $5 \times \#\, regimes + 1$. The parameters $\delta$, $\eta_\Theta$, and $\eta_\Gamma$ of equations (1) and (2) are respectively set to 3, $10^{-6}$, and $10^{-6}$. These values are kept unchanged in the rest of the paper. We also use the priors presented in Table 1.

Table 2 displays the results when breaks are in $\omega$ and Table 3 when they are in $\beta$. It can be seen from Table 2 that the D-DREAM algorithm selects the right model most of the time. However, focusing on the DGPs that exhibit a structural break, the smaller the difference between the $\omega$ parameters of the two regimes or the greater the persistence of the volatility dynamics, the more difficult it is to favour the model with the structural break. For instance, for a relatively large parameter change of 0.4 ($\omega_1 = 0.1$ and $\omega_2 = 0.5$) and a persistence ($\alpha + \beta + \gamma/2$) of 0.9 (unchanged from regime 1 to regime 2) the MLL of the model with one structural break is always found to be larger. If the parameter change is reduced to 0.2 ($\omega_2 = 0.3$), the model with one change point is chosen only 51 times (out of 100). If, instead of modifying the parameter change, the persistence is increased to 0.95, the model with change-point is chosen only 77 times. It is also worth noting that break dates are more accurately and precisely estimated if the change in parameter is great or persistence is low. For example, for a change in $\omega$ of 0.4 and a persistence of 0.8, the average posterior mean of the break date is 501.42 (recall that the true break date is located at observation 500) and the 97.5%-2.5% percentile range is 27.32 (days), which is rather accurate and precise. On the contrary, for a change in $\omega$ of 0.2 and a persistence of 0.95, the average posterior mean of the break date is 490.62 and the 97.5%-2.5% percentile range is 161.86, which is much less accurate.

Results regarding changes in the $\beta$ parameter need careful interpretation. It can be seen from Table 3 that small variations in the $\beta$ parameter are more difficult to detect when $\beta$ is small. For instance, when $\beta$ is 0.6 and increases to 0.7 in regime 2 the model without structural break is preferred on average. This can be seen from the average of the logarithm of the Bayes factors which is negative (-4.46). On the other hand, when $\beta$ is 0.8 and increases to 0.9 the average of the logarithm of the Bayes factor is 3.70 and the model with a structural break is chosen most of the time (82 times out of 100). In order to understand this result, notice that the parameter $\beta$ has a nonlinear impact on the unconditional variance as well as on the autocorrelation function of the conditional variance. Consequently, for an equal increase, the larger the original beta ($\beta_1$), the larger the impact on the unconditional variance and on the autocorrelation of the conditional variance, and the easier it is to detect a change in regime.

[4]The MLLs of pure change-point models were computed via Bridge sampling, as for the rest of the paper.



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| | Persistence | $\omega_2$ | | | |
|---|---|---|---|---|---|
| | | 0.1 (no break) | 0.3 | 0.5 | 0.7 |
| | A | # models with one break selected | | | |
| | 0.7 | 0 | 100 | 100 | 99 |
| | 0.8 | 0 | 93 | 100 | 100 |
| | 0.9 | 1 | 51 | 100 | 98 |
| | 0.95 | 0 | 34 | 77 | 94 |
| | B | Average log BF | | | |
| | 0.7 | -8.75 | 10.75 | 18.05 | 21.10 |
| | 0.8 | -8.63 | 7.29 | 14.45 | 16.94 |
| | 0.9 | -6.63 | 1.18 | 9.58 | 12.17 |
| | 0.95 | -6.50 | -1.04 | 4.01 | 7.18 |
| $\omega_1 = 0.1$ | C | Average posterior break mean | | | |
| | 0.7 | 604.94 | 501.61 | 500.62 | 500.38 |
| | 0.8 | 675.05 | 500.21 | 501.42 | 501.31 |
| | 0.9 | 589.03 | 501.47 | 503.98 | 500.47 |
| | 0.95 | 508.81 | 490.62 | 505.53 | 500.17 |
| | D | Avg. 95% post. range break range | | | |
| | 0.7 | 490.35 | 46.31 | 18.70 | 13.87 |
| | 0.8 | 544.07 | 56.01 | 27.32 | 24.66 |
| | 0.9 | 334.63 | 129.32 | 35.92 | 26.13 |
| | 0.95 | 241.36 | 161.86 | 71.23 | 43.37 |

Table 2: Monte Carlo Simulations 1: Breaks in $\omega$. The table shows the performance of the D-DREAM algorithm after 100 simulations of 16 DGPs. Breaks in the parameter $\omega$ happen at observation 501 (half of series lengths). The persistence ($\alpha + \beta + \gamma/2$) is shown in the second column and is the same in both regimes. Panel A displays the number of times the MLL of the model with one break, computed via Bridge sampling, is greater than the MLL of the model without change-point. Panel B shows the averages of the logarithms of the Bayes Factors (BF), that is, the MLL of the model with one break minus the MLL of the model without change-point. Panel C displays the average posterior means of the break points. Panel D shows the average differences between the 97.5% percentile and the 2.5% percentile (in days).



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| | $\beta_1$ | $\beta_2$ | | | |
|---|---|---|---|---|---|
| | | 0.6 | 0.7 | 0.8 | 0.9 |
| | A | # Models with one break selected | | | |
| | 0.6 | 1 | 0 | 94 | 100 |
| | 0.7 | 2 | 2 | 80 | 100 |
| | 0.8 | 90 | 57 | 0 | 82 |
| | 0.9 | 99 | 96 | 78 | 6 |
| | B | Average log BF | | | |
| $\omega = 0.1$ | 0.6 | -6.65 | -4.46 | 5.46 | 13.91 |
| $\alpha = 0.05$ | 0.7 | -4.31 | -6.05 | 1.51 | 10.22 |
| $\gamma = 0.05$ | 0.8 | 6.86 | 1.41 | -5.48 | 3.70 |
| | 0.9 | 14.33 | 9.76 | 4.05 | -3.46 |
| | C | Average posterior break mean | | | |
| | 0.6 | 547.92 | 538.08 | 505.77 | 501.61 |
| | 0.7 | 624.87 | 650.45 | 508.98 | 503.82 |
| | 0.8 | 497.25 | 454.01 | 507.55 | 489.27 |
| | 0.9 | 498.13 | 497.82 | 494.72 | 631.99 |
| | D | Avg. 95% post. range break range | | | |
| | 0.6 | 523.45 | 254.90 | 71.50 | 24.48 |
| | 0.7 | 271.50 | 432.05 | 181.24 | 36.25 |
| | 0.8 | 72.38 | 183.76 | 479.43 | 90.94 |
| | 0.9 | 15.25 | 25.23 | 62.20 | 281.23 |

Table 3: Monte Carlo Simulations 1: Breaks in $\beta$. The table shows the performance of the D-DREAM algorithm after 100 simulations of 16 DGPs. Break in the parameter $\beta$ happen at observation 501 (half of series lengths). The parameters $\omega$, $\alpha$, and $\gamma$ are set to 0.1, 0.05, and 0.05, respectively and are the same in both regimes. Panel A displays the number of times the MLL of the model with one break, computed via Bridge sampling, is greater than the MLL of the model without change-point. Panel B shows the averages of the logarithms of the Bayes Factors (BF), that is, the MLL of the model with one break minus the MLL of the model without change-point. Panel C displays the average posterior means of the break points. Panel D shows the average differences between the 97.5% percentile and the 2.5% percentile (in days).

*5.2. Monte Carlo Simulations 2: Recurrent Regimes*

We illustrate further the flexibility of the D-DREAM algorithm by showing that it can efficiently deal with recurrent regimes, as explained in section 3. We simulate time series of 4000 observations according to a CP-GJR-GARCH(1,1) model with three change points and one recurrent regime. The DGP is displayed in Table 4. The values of the different parameters are chosen to be representative of financial time series and to switch from quiet periods to more volatile ones.

The Monte Carlo study consists of 100 simulated series. For each simulated series we compute the MLL of CP-GJR-GARCH(1,1) models with $K = 0, 1, ..., 4$, considering all



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| Parameters | Regime 1 | Regime 2 and 4 | Regime 3 |
|---|---|---|---|
| $\mu$ | 0 | 0 | 0 |
| $\omega$ | 0.400 | 0.300 | 0.150 |
| $\alpha$ | 0.125 | 0.050 | 0.075 |
| $\gamma$ | 0.050 | 0.100 | 0.050 |
| $\beta$ | 0.550 | 0.850 | 0.650 |
| Break Dates | 1000 | 2000 — | 3000 |

Table 4: DGP With Recurrent Regime.

possible regime combinations. The result is striking: in 100 simulations the right model is selected 96 times according to the MLL criterion (three breaks and the last regime equal to the second).[5] The average of the posterior means of the model parameters and break dates of the right specification are reported in Table 5. The values are close to the true values of the DGP and the structural break dates are particularly well estimated.

| Parameters | Regime 1 | Regime 2 and 4 | Regime 3 |
|---|---|---|---|
| $\mu$ | -0.00 | 0.00 | -0.00 |
| | (0.04) | (0.05) | (0.03) |
| $\omega$ | 0.54 | 0.35 | 0.23 |
| | (0.14) | (0.10) | (0.14) |
| $\alpha$ | 0.13 | 0.05 | 0.09 |
| | (0.05) | (0.01) | (0.04) |
| $\gamma$ | 0.07 | 0.11 | 0.06 |
| | (0.07) | (0.03) | (0.06) |
| $\beta$ | 0.43 | 0.84 | 0.50 |
| | (0.12) | (0.03) | (0.24) |
| Break Dates | 1000.79 | 1999.16 — | 3000.19 |
| | (16.18) | (6.76) — | (4.40) |

Table 5: Averages of Posterior Means of Right Specification. The table displays the averages, over 100 simulations, of the posterior means of the model parameters and break dates of the right specification. Standard deviations of the 96 posterior means are reported between parentheses. This table must be compared with Table 4.

[5]Obviously, the exercise of finding breaks in a Monte Carlo simulation can be made as hard as desired. With another DGP with "smaller" breaks and regimes with an unconditional standard deviation evolving from 0.91 to 1.73 with an intermediate state equal to 1.20 *and* a persistence that switches from 0.7, to 0.82, to 0.95, we still find that the correct recurrence pattern is chosen 42 times out of 100. Moreover, the number of regimes is never over-estimated and 98 of the selected models contain structural breaks.



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In addition, we show that the two techniques we use to compute the MLL, namely Bridge sampling and importance sampling, give equivalent results. We compare the MLL estimators by simulating one time series and by conducting 100 MCMC simulations along with MLL estimations. Table 6 displays for all $K$ the differences between the averages (over the 100 estimations) of the MLLs. It appears that the differences grow with the number of regimes but remain small and insignificant. This equivalence is largely due to the bell shape of the posterior distributions of the parameters.

| | $K$ | | | | |
|---|---|---|---|---|---|
| | 0 | 1 | 2 | 3 | 4 |
| MLL averages differences | 0.03 | -0.15 | -0.03 | 1.97 | 4.94 |
| (std. dev.) | (0.09) | (0.62) | (0.87) | (1.43) | (3.76) |
| | MLL Importance sampling | | | | |
| MLL average | -7349.48 | -7344.93 | -7346.93 | -7328.59 | -7339.85 |
| (std. dev.) | (0.09) | (0.87) | (0.78) | (1.64) | (4.14) |
| | MLL Bridge sampling | | | | |
| MLL average | -7349.45 | -7345.07 | -7346.96 | -7326.62 | -7334.91 |
| (std. dev.) | (0.05) | (0.64) | (0.78) | (0.76) | (2.75) |

Table 6: Bridge Sampling vs Importance Sampling: MLL Estimates. The table displays the differences between the averages of the MLL estimation via Bridge sampling and Importance sampling. Standard deviations from the 100 simulations are in parentheses.

*5.3. Comparison with Other Algorithms*

The break date sampling method of the D-DREAM algorithm relies on Metropolis steps whereas the single-move (Bauwens et al., 2010, BPR) and the PMCMC (Bauwens et al., 2013) algorithms are Gibbs samplers applied to the state variables. Although sampling from the conditional distributions is practical (in the case of the single-move) and usually more efficient in terms of mixing properties than a Metropolis algorithm, we argue that the D-DREAM algorithm dominates the single-move and the PMCMC algorithms.

To support our view, we have carried out a comparison exercise on the S&P 500 daily return series with a fixed number of regimes equal to 5 (data from June 20, 1991, to April 25, 2011, 5000 observations, see section 6). All the algorithms have been run with the same initial point (and additional initial points for the D-DREAM algorithm which runs multiple chains in parallel) and 5000 posterior draws were retained after the convergence of the Markov chains. The three algorithms provide similar results in terms of posterior means of break dates (reported in Section 6.1) but we focus here on the mixing properties. Table 7 documents the autocorrelation times,[6] computed by batch means, of each break

[6]Defined as $1 + 2\sum_{i=1}^{\infty} \rho_i$ where $\rho_i$ is the autocorrelation of order $i$ of the posterior draws of the break



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date parameter (see Geyer, 1992). The posterior draws of the BPR algorithm are highly autocorrelated, while the PMCMC and D-DREAM algorithms perform much better.

| Regime | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| BPR | 241.17 | 310.95 | 315.52 | 246.59 |
| PMCMC | **21.30** | 118.29 | 79.93 | **4.43** |
| D-DREAM | 72.16 | **26.54** | **46.86** | 47.63 |

Table 7: Autocorrelation Time. The table displays the autocorrelation times computed by batch means of the break date parameters for the three different algorithms.

The D-DREAM algorithm is, however, much faster than the PMCMC. To show this, Table 8 reports the elapsed time in minutes to obtain 5000 posterior draws as well as the time in seconds for sampling an uncorrelated draw taking into account only the lowest autocorrelation time. In particular, it took a bit more than 6 minutes for the D-DREAM algorithm to complete the sampling, while it took more than 6 hours for the PMCMC alternative. Moreover, sampling an uncorrelated draw takes 2 seconds for the D-DREAM algorithm whereas it takes respectively 10 and 20 seconds for the BPR and the PMCMC competitors. The D-DREAM algorithm also dominates the other ones when using the highest autocorrelation times or their mean.

| | Elapsed time (minutes) | Time for one uncorrelated draw (seconds) |
|---|---|---|
| BPR | **3.5** | 10.13 |
| PMCMC | 372.8 | 19.82 |
| D-DREAM | 6.25 | **1.99** |

Table 8: Elapsed Time and Time for Uncorrelated Draw. The table displays the elapsed times (in minutes) to obtain 5000 posterior draws and times (in seconds) for sampling one uncorrelated posterior draw for the three different algorithms.

## 6. Empirical Applications to Financial Time Series

This section documents results of estimation of the CP-GJR-GARCH(1,1) model defined in section 2, and other related models, by the D-DREAM algorithm on seven financial time series. We start by focusing on the S&P 500 index with the CP-GJR-GARCH(1,1) and then briefly comment the corresponding results for six other time series in subsection 6.2. For the same seven series, in subsection 6.3, we estimate another kind of regime-switching GARCH model proposed by Haas et al. (2004), to illustrate the versatility of

dates. The autocorrelation time is interpretable as the number of draws from the MCMC needed to obtain one new uncorrelated draw from the posterior distribution.



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the D-DREAM algorithm and to compare both types of models by the MLL criterion. Finally, in subsection 6.4, we replace the assumption of normality of the innovations of the CP-GJR-GARCH(1,1) model defined in section 2 by the assumption that the distribution of the innovations is Student-$t$, in order to check the robustness of the inferences for the same series to this type of assumption.

*6.1. S&P 500*

Our time series for the (weighted) Standard & Poor's 500 index consists of daily returns from June 20, 1991 to April 25, 2011 (5000 observations) and was downloaded from Datastream.

| # Regimes | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $\lambda$ | $\{1\}$ | $\{1,2\}$ | $\{1,2,3\}$ | $\{1,2,1,2\}$ | $\{1,2,3,2,3\}$ | $\{1,2,1,4,2,4\}$ |
| MLL | -6726.66 | -6730.06 | -6725.00 | -6706.45 | **-6705.30** | -6709.19 |

Table 9: S&P 500 Highest MLLs. The table displays the highest MLLs among the possible regime combinations for each number of regimes considered. The row '$\lambda$' indicates the regime combination that yields the highest MLL for the number of regimes in the first row. The maximum highest MLL is in bold.

Focusing first on the number of regimes and their recurrence, it is shown in Table 9 that the MLL of a CP-GJR-GARCH(1,1) model is maximized for a specification with four structural breaks and with the fourth regime equal to the second and the fifth equal to the third. The posterior means of the break dates are January 30, 1992, November 20, 1996, September 17, 2003, and June 5, 2007. Notice however that the specification with four regimes is only slightly dominated by the preferred one (the difference in MLL values being less than one), while the other specifications are clearly dominated. Actually, the specification with five regimes differs from the model with four regimes by the presence of a short-lived regime of 156 days at the beginning of the sample in the former.

These results seem plausible in light of Figure 2 that plots the S&P 500 return series, along with the posterior means and 95% confidence bands of the structural break dates. In particular, according to the figure, the volatility of the series looks indeed similar and low between February 1992 and November 1996, and between September 2003 and June 2007. It is perhaps surprising that the regime of high volatility of the last financial crisis (after May 2007) is not different from the corresponding regime between November 1996 and September 2003. Instead, we find that the dynamics of the S&P 500 after June 2007 are comparable to the dynamics of the series at the end of the last century and beginning of the $21^{st}$ century. However the latter period was marked by events like the Russian default, the LTCM collapse, the "dot-com" bubble, and the attack on the World Trade Center. These events contributed to a large extent to add volatility in stock prices and therefore, given our priors, the D-DREAM algorithm does not find it worth creating a new regime for the financial crisis.



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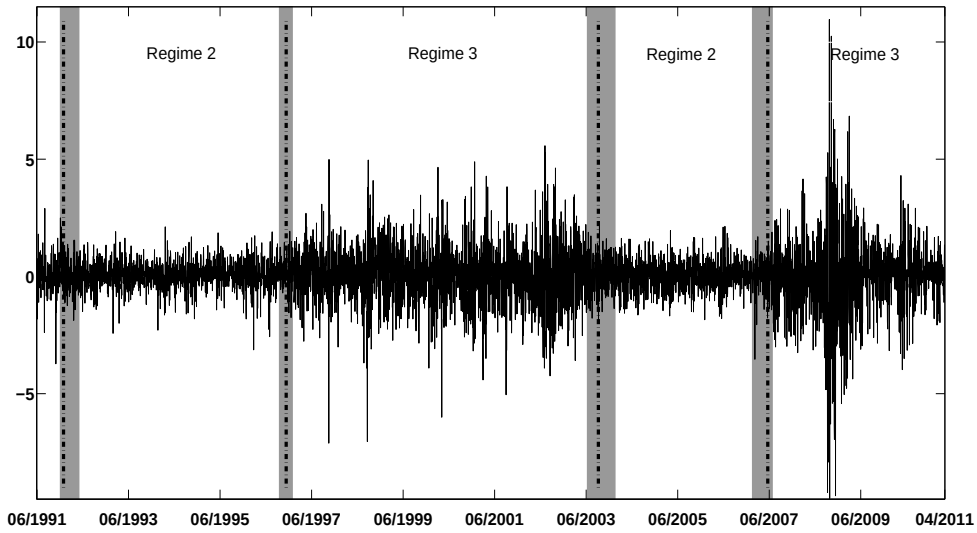



Figure 2: S&P 500 returns (in percentage) with Posterior Means of Structural Breaks (dashed lines) and 95% Confidence Intervals (shaded areas).

| Parameters | Regime 1 | Regimes 2 & 4 | Regimes 3 & 5 | |
|---|---|---|---|---|
| $\mu$ | 0.064 | 0.037 | -0.001 | |
| | (0.063) | (0.012) | (0.019) | |
| $\omega$ | 0.447 | 0.037 | 0.045 | |
| | (0.110) | (0.007) | (0.007) | |
| $\alpha$ | -0.010 | -0.045 | -0.027 | |
| | (0.069) | (0.010) | (0.007) | |
| $\gamma$ | 0.218 | 0.162 | 0.189 | |
| | (0.165) | (0.021) | (0.015) | |
| $\beta$ | 0.283 | 0.869 | 0.910 | |
| | (0.137) | (0.019) | (0.011) | |
| Break Dates | 30-Jan-92 | 20-Nov-96 | 30-Jun-07 | 19-Sep-03 | — |
| | (29.47) | (17.91) | (25.16) | (36.40) | — |
| Durations in days | 156 | 1217 | 929 | 1717 | 982 |

Table 10: S&P 500 Posterior Results. The table displays the posterior means and standard deviations (in parentheses) of the parameters and break dates of the best CP-GJR-GARCH(1,1) specification. Durations of regimes, expressed in days, are computed with the posterior means of break dates as landmarks.

Table 10 shows the posterior means of the GARCH parameters and break dates of the



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best specification according to the MLL criterion. As could be expected, the series switches between quiet periods (regimes 2 and 4, with a positive mean return, low unconditional standard deviation of 0.62%, and persistence of 0.91) and turbulent episodes (regimes 3 and 5, with a negative mean return, unconditional standard deviation of 1.41% and persistence of 0.98). Notice also that in all regimes the posterior means of the parameter $\gamma$ of the leverage effect are strongly positive while the posterior means of the $\alpha$ parameter are slightly negative. The large posterior standard deviations in the results for regime 1 reflect the brevity of the available sample for that regime.

*6.2. Other Financial Series*

We apply the algorithm with a CP-GJR-GARCH(1,1) model to six other financial time series of daily returns spanning the same time period as the S&P500 and downloaded from the same source. The series are the (weighted) Dow Jones Industrial Average (DJIA) index, the (weighted) New York Stock Exchange (NYSE) composite index, gold prices ("Gold", Gold Bullion London Bullion Market USD/Troy Ounce), the West Face Capital Inc. Base Metals sub-index ("Metals"), share prices for Boeing Co. (BA) and JP Morgan Chase & Co. (JPM). For share prices, we do not consider that dividends are reinvested in shares.

| Series | Number of Breaks $K$ | Regime Combination $\lambda$ |
|---|---|---|
| DJIA | 3 | {1,2,1,2} |
| Break Dates: | 15-Nov-96, 08-Aug-03, 07-Jun-07 | |
| NYSE | 4 | {1,2,3,2,3} |
| Break Dates: | 28-Jan-92, 26-Dec-96, 08-Sep-03, 17-Jan-07 | |
| Gold | 3 | {1,2,3,1} |
| Break Dates: | 22-Nov-96, 27-Aug-01, 14-Nov-01 | |
| Metals | 3 | {1,2,3,4} |
| Break Dates: | 19-Jul-93, 27-May-03, 11-Nov-06 | |
| Boeing | 3 | {1,2,3,1} |
| Break Dates: | 04-Jun-98, 01-Dec-98, 29-Aug-02 | |
| JP Morgan | 2 | {1,2,1} |
| Break Dates: | 08-Sep-03, 13-Apr-07 | |

Table 11: CP-GJR-GARCH(1,1) Specifications with Highest MLL. The table displays the best specification for each series according to the MLL criterion, and the corresponding break dates.

Table 11 displays the main characteristics of the best specification ($K$, $\lambda$, break dates) for each series according to the MLL criterion. We find that all series contain structural breaks and that five of them entail at least one recurrent regime. This finding should be placed in perspective with the results of Bauwens et al. (2013) who apply their PMCMC algorithm to similar series and find that Markov-switching models fit the data better than



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pure change-point models. Notice that the DJIA and NYSE time series are similar to the S&P500 and thus, yield similar break points compared with those of the S&P500. Another interesting finding is that the leverage parameter $\gamma$ is significantly negative in three regimes for the Gold series, and in two regimes for the Metals series (posterior results for all series are presented in Appendix C). This corroborates the results of Carpantier and Dufays (2012) who fit a GJR-GARCH model to many commodity series and almost systematically find an inverse leverage effect. Hence, it seems that their results are robust to the inclusion of structural breaks in econometric models.

*6.3. Comparison with the Model of Haas, Mittnik, and Paolella (2004)*

We apply the D-DREAM algorithm to the seven series of financial returns using a change-point model that differs from the model used in the two previous subsections. That alternative model is a change-point version of the Markov-switching GARCH model proposed in Haas et al. (2004). We name it CP-HMP. In that model, when a change-point takes effect, the conditional variance after the break (thus of the new dynamics) depends on the past shocks and conditional variances induced by the new dynamics, as if it had been active from the beginning of the sample. This modification of the CP-GJR-GARCH model presented in section 2 eliminates the path dependence problem. Accordingly, the CP-GJR-GARCH model differs from the CP-HMP model especially when a structural break occurs. Since we only detect a small number of breaks in the seven series used in this study, we can expect that the two models lead to similar results.

To be more specific, let us consider the one-to-one mapping between the break dates $\{\tau_0, ..., \tau_{K+1}\}$ and the state vector $S_T = \{s_1, s_2, ..., s_T\}$ in which the variable $s_t$ is discrete, its value (in $[1, K+1]$) indicating the regime operating at time $t$. The difference between the CP-HMP and the CP-GJR-GARCH models can be highlighted by expressing them as ARCH($\infty$) processes:

$$\begin{aligned}
\text{CP-GJR-GARCH:}\quad \sigma_t^2 = \sigma_0^2 \prod_{j=1}^{t} \beta_{s_j} + \sum_{k=1}^{t-1} [(\omega_{s_k} + (\alpha_{s_k} + \gamma_{s_k} \mathbb{1}_{\epsilon_{k-1}<0}) \epsilon_{k-1}^2) \prod_{j=k}^{t-1} \beta_{s_{j+1}}] \\
+ \omega_{s_t} + (\alpha_{s_t} + \gamma_{s_t} \mathbb{1}_{\epsilon_{t-1}<0}) \epsilon_{t-1}^2 \\
\text{CP-HMP:}\quad \sigma_{t,i}^2 = \sigma_0^2 \beta_i^t + \sum_{k=1}^{t} (\omega_i + (\alpha_i + \gamma_i \mathbb{1}_{\epsilon_{k-1,i}<0}) \epsilon_{k-1,i}^2) \beta_i^{t-k}, \\
i = 1, ..., K+1,
\end{aligned}$$

where $i$ is the label of the GJR-GARCH process that is operating at time $t$.

These formulas highlight the strong similarity of the volatility dynamics of the two models. We observe that the impact of past variances on the current one geometrically decays, bringing the two dynamics closer when the lengths of regimes increase. Furthermore, the two volatility equations are piecewise linear functions of past squared shocks on the financial return. These innovations differ mainly when, again, the processes face a structural break, which scarcely occurs in change-point configurations. When a break



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happens, the CP-HMP model adjusts the previous shocks according to the new dynamics while the CP-GJR-GARCH model keeps them unchanged.

Posterior results for both change-point models have been generated by the D-DREAM algorithm under the prior specification documented in Table 1, except for the persistence parameter $\beta$ that has to stay below unity, as required for the HMP model. For these comparisons, we consider pure CP models, that is models without recurrent regimes.

Table 12 documents the posterior means and standard deviations of the break date parameters for the S&P 500 series. We observe that the estimated break dates (and their standard deviations) are very similar, meaning that the two models display closely related volatility dynamics. Marginal Log-Likelihoods (MLLs) are also quite close and indicate that two breaks are selected in each model.

| Regime | 2 | 3 | 4 | 5 | MLL |
|---|---|---|---|---|---|
| CP-HMP | — | | | | -6726.66 |
| CP-GJR-GARCH | — | | | | -6726.66 |
| CP-HMP | 29/04/1996 (145.72) | — | | | -6729.25 |
| CP-GJR-GARCH | 28/06/1996 (123.39) | — | | | -6730.06 |
| CP-HMP | 02/01/1997 (37.11) | 21/08/2000 (136.47) | — | | **-6725.36** |
| CP-GJR-GARCH | 20/12/1996 (30.74) | 05/09/2000 (147.34) | — | | **-6725.00** |
| CP-HMP | 21/11/1996 (51.99) | 03/09/2003 (26.07) | 27/06/2007 (17.90) | — | -6731.52 |
| CP-GJR-GARCH | 05/12/1996 (22.93) | 20/08/2003 (27.51) | 06/06/2007 (24.82) | — | -6732.36 |
| CP-HMP | 24/02/1992 (40.17) | 10/12/1996 (16.66) | 04/09/2003 (25.04) | 25/06/2007 (20.18) | -6731.22 |
| CP-GJR-GARCH | 07/02/1992 (38.81) | 29/11/1996 (15.76) | 15/08/2003 (26.63) | 07/06/2007 (22.13) | -6732.35 |

Table 12: Posterior Means and Standard Deviations (in days) of Break Dates of the CP-HMP and the CP-GJR-GARCH(1,1) Models (S&P 500 series). Each date indicates the beginning of the corresponding regime in the first row of the table. MLLs are reported in the last column. Bold values indicate the largest MLL for each model.

In Table 13, we report the MLLs for the six other financial series studied in subsection 6.2. The MLLs of the two models are often very close to each other, and therefore there is no clear evidence that one model systematically performs better than the other. In addition to showing that the two competing models are providing similar break dates, the



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results of this comparison illustrate the possibility to use the D-DREAM algorithm with different classes of models. Further applications of this type are left for future research.

| #Regime | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| | | | DJIA | | |
| CP-HMP | -6637.60 | **-6635.59** | *-6635.67* | *-6641.66* | *-6646.64* |
| CP-GJR-GARCH | -6637.60 | -6636.27 | -6636.08 | -6641.69 | -6649.29 |
| | | | NYSE | | |
| CP-HMP | -6414.05 | **-6405.37** | -6411.36 | -6411.11 | -6417.17 |
| CP-GJR-GARCH | -6414.05 | -6406.81 | *-6408.7* | *-6408.46* | *-6416.74* |
| | | | Gold | | |
| CP-HMP | -6001.14 | -5997.00 | -5962.83 | -5960.42 | *-5978.6* |
| CP-GJR-GARCH | -6001.14 | *-5995.25* | **-5958.24** | *-5959.81* | -5984.24 |
| | | | Metals | | |
| CP-HMP | -7795.34 | -7789.53 | -7780.74 | -7772.45 | -7777.32 |
| CP-GJR-GARCH | -7795.34 | *-7788.46* | *-7779.73* | **-7772.37** | *-7775.96* |
| | | | Boeing | | |
| CP-HMP | -10071.12 | -10081.51 | *-10048.94* | **-10048.59** | -10062.08 |
| CP-GJR-GARCH | -10071.12 | *-10079.21* | -10051.5 | -10049.77 | *-10056.01* |
| | | | JP Morgan | | |
| CP-HMP | -10477.67 | -10484.4 | -10485.35 | -10490.03 | *-10494.07* |
| CP-GJR-GARCH | -10477.67 | *-10482.00* | *-10483.89* | *-10488.96* | -10494.48 |

Table 13: MLL of the CP-HMP and the CP-GJR-GARCH Models (for six financial time series). Values in italics indicate the highest MLL between the two models for each series and number of regimes greater than 1. Bold values indicate the highest MLL for each series.

### *6.4. Comparison with Student-t Distributions for the Innovations*

The GJR-GARCH dynamics and the mixture of Normals resulting from the different regimes partly account for the fat tails present in the unconditional distribution of the financial series studied in the previous subsections. However, fat tails are usually better captured by assuming a Student-$t$ distribution for the innovations. Applying the D-DREAM algorithm with the latter distribution, we find structural breaks in the CP-GJR-GARCH models. However, according to the marginal likelihood criterion, the selected GJR-GARCH(1,1) models exhibit fewer breaks than with a Normal distribution, and, for some series, no break at all. In particular, we do not find structural breaks in the S&P 500 series, as shown in Table 14. The posterior results for the single regime model parameters are displayed in Table 15. The posterior mean of the number of degrees of freedom (dof)



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is equal to 7.74, reflecting the fat tails of the series. Table 16 shows a summary of the results for the other six series. We find one break for the series NYSE and Gold, two for Metals and Boeing, and no break for DJIA and JP Morgan. The series NYSE, Gold, and Metal follow a pure change-point model, while the first regime is recurrent for Boeing. The posterior means of the model parameters for these six series are given in Appendix D. In cases in which breaks occur, the degrees-of-freedom parameter changes significantly (except between regimes 2 and 3 for Metals), showing the potential usefulness of the break model with Student-$t$ innovations.

| # Regimes | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $\lambda$ | $\{1\}$ | $\{1,2\}$ | $\{1,2,1\}$ | $\{1,2,3,1\}$ | $\{1,2,1,3,1\}$ |
| MLL | **-6640.74** | -6645.61 | -6653.96 | -6643.75 | -6653.85 |

Table 14: S&P 500 Highest MLLs (Model with Student-$t$ Distribution). The table displays the highest MLLs among the possible regime combinations for each number of regimes considered. The row '$\lambda$' indicates the regime combination that yields the highest MLL for each number of regimes. The maximum highest MLL is in bold.

| Posterior | $\mu$ | $\omega$ | $\alpha$ | $\gamma$ | $\beta$ | dof |
|---|---|---|---|---|---|---|
| Mean | 0.042 | 0.007 | -0.006 | 0.097 | 0.933 | 7.742 |
| St. dev. | (0.010) | (0.001) | (0.004) | (0.009) | (0.006) | (0.781) |

Table 15: S&P 500 Posterior Results (Model with Student-$t$ Distribution). The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates of the best CP-GJR-GARCH(1,1) specification (i.e. the single regime model).

The reason why we find less structural breaks according to the MLL criterion assuming a Student-$t$ distribution for the innovations can be explained. It is related to the penalisation that we impose through the prior density each time an additional regime is considered, and to the data likelihood improvement brought by the additional regime. Let us assume that the third term (the log posterior) in the local formula of the MLL, $\log f(Y_T) = \log f(Y_T|\Theta,\Gamma) + \log p(\Theta,\Gamma) - \log \pi(\Theta,\Gamma|Y_T)$, remains unchanged. Then, in order to favour an additional regime, the extra set of parameters brought along by the additional regime should increase the likelihood of the data more than it penalises the marginal likelihood through the prior density. This is more difficult to achieve with Student-$t$ innovations. To see this, notice that the Student-$t$ distribution (with a low dof) accommodates much more easily than the Normal density the passage from a quiet period to a highly volatile period thanks to its fat tails. Consequently, for the same prior as with Normal innovations, that is, with the same penalisation for an additional regime, a structural break needs to be stronger in the case of Student-$t$ innovations for the MLL to find it worth adding the whole set of parameters of a new regime.



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| Series | Number of Breaks $K$ | Regime Combination $\lambda$ |
|---|---|---|
| DJIA | 0 | {1} |
| Break Dates: | — | |
| NYSE | 1 | {1,2} |
| Break Dates: | 10-Sept-96 | |
| Gold | 1 | {1,2} |
| Break Dates: | 7-Feb-02 | |
| Metals | 2 | {1,2,3} |
| Break Dates: | 9-Jul-93, 15-Aug-03 | |
| Boeing | 2 | {1,2,1} |
| Break Dates: | 1-May-97, 7-Feb-02 | |
| JP Morgan | 0 | {1} |
| Break Dates: | — | |

Table 16: CP-GJR-GARCH(1,1) Specifications with Student-$t$ Distribution with Highest MLL. The table displays the best specification for each series according to the MLL criterion, and the corresponding break dates.

Two alternatives could be considered to let the MLL criterion select more easily a model with more breaks (with a Student-$t$ or any other distribution). First, we could allow some, but not all, of the parameters of the GJR-GARCH(1,1) model to change, so as to reduce the number of parameters of the specification with several regimes and thus reduce the marginal cost of an additional regime. Second, we could also help the MLL select models with more breaks by using some a priori information we might have on the location of break dates. Since the goal of the paper is to introduce the D-DREAM algorithm and show that direct sampling of the break dates can be performed with a Metropolis algorithm, even in the case of path dependent processes, we decided not to investigate this issue further.

## 7. Forecasting Comparisons

Structural break models do not necessarily outperform models with fixed coefficients regarding their forecasting ability. They indeed need enough observations once a new regime is detected to estimate the parameters accurately. If the estimates are imprecise, structural break models can actually do worse at forecasting than fixed parameter models based on a rolling or expanding window. Therefore, we compare the forecasting accuracy of GJR-GARCH and CP-GJR-GARCH models using the predictive marginal likelihood (PML) criterion. The idea of the PML criterion is to compare the marginal likelihoods of observation $y_{t+1}$ of several models after estimating these models with the information available up to time $t$, and to select the model exhibiting the highest marginal likelihood. Hence, we need to evaluate $f(y_{t+1}|Y_t)$ for each model. This can be done by averaging the



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likelihoods $f(y_{t+1}|\Theta_{g_1}, \Gamma_{g_1}, Y_t)$, where $\Theta_{g_1}$ and $\Gamma_{g_1}$ denote the $g_1^{th}$ of $G_1$ posterior draws of the model parameters and break dates.

We perform a forecasting exercise at horizons $h = \{1, 2, 5, 10, 20, 40, 50\}$ based on the seven series used in Section 6. We start to forecast the series in the middle of the sample (observation 2500) and we add fifty observations to the estimation window before computing the next forecast, resulting in 49 forecast evaluations since all series contain 5000 observations. At each of the 49 evaluations, we run the D-DREAM algorithm with a CP-GJR-GARCH(1,1) model for $K = \{0, 1, 2, ..., 6\}$, with possibly recurrent regimes (for $K \geq 2$), and without recurrence (for $K \geq 0$). Then we compare the forecast accuracy of six different classes of CP-GJR-GARCH(1,1) models: (1) a fixed parameter model ($K = 0$) with Normal innovations; (2) the class of models allowing for structural breaks and recurrent regimes with Normal innovations; (3) the class of models allowing for structural breaks only (no recurrent regimes) with Normal innovations; (4), (5), and (6) the same specifications as (1), (2), and (3), respectively, but with Student-$t$ innovations. For the model classes 2, 3, 5, and 6, we forecast the series with the best specification according to the MLL criterion (the MLL selects $K$ and $\lambda$ for the classes 2 and 5, and only $K$ for the classes 3 and 6).

Tables 17 and 18 show the average log-PMLs, over all 49 forecasts, of the six classes of models for all forecast horizons. First of all, the results show that the models with fixed parameters dominate the change-point models: on a total of 49 cases (7 forecast horizons $\times$ 7 series), the fixed parameter models (with Normal or Student-$t$ innovations) beat the structural break models (with or without recurrent regimes) 35 times (73%). Secondly, the models with Student-$t$ innovations outperform the models with Normal innovations: among the 35 cases mentioned above, 33 correspond to models with Student-$t$ innovations (67% of all cases), and globally there are only four cases (for the Gold series, at horizons 1, 2, and 5, and the Boeing series at horizon 2) where models with Normal innovations win. Thirdly, for three series that present change points when the models are estimated on the whole sample with Student-$t$ distributions, namely Gold, Metals and Boeing, CP models with Student-$t$ distributions have the best PML scores 11 times (52%).

We wish however to mitigate a bit these results. Firstly, it is the case that for about 29% of the cases, a change-point model provides a higher PML than a fixed parameter model. Thus change-point models can provide value added in some cases. Secondly, we can see from Tables 17 and 18 that log-PML values are often very close to each other across models, given the values of the standard deviations.

It is worth mentioning that the finding that fixed parameter models perform as well as change-point models regarding their forecasting accuracy is not new. In a forecasting exercise on macroeconomic series, Bauwens et al. (2014) find that fixed parameter autoregressive (AR) models often beat structural break AR models and outperform them on average according to the PML criterion. Interestingly, they find the opposite result according to the root mean squared error (RMSE) criterion. Notice that in the context of volatility models without a proper modelling of the mean equation, one cannot use a criterion like the RMSE because volatilities are not directly observed. Finally, since we work with models for the volatility dynamics, with static mean equations, forecast improve-



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ments are mainly obtained from the forecast conditional variances, which makes forecast likelihoods harder to enhance by taking structural breaks into account.

## 8. Conclusions

Change points are generally accounted for by using the method of Chib (1996), which introduces a hidden state variable that indicates the regimes which the observations belong to. However, this method is computationally very demanding, or infeasible, in models subject to the path dependence problem. In this paper, we introduce a feasible and computationally efficient MCMC algorithm by treating break dates directly as parameters, thus avoiding the path dependence problem, and we prove the convergence of the algorithm. The number of breaks is determined thanks to the marginal likelihood criterion. Moreover, we allow for the regimes to be potentially active in several distinct periods throughout the sample.

Focusing on two kinds of change-point GARCH models, we illustrate the efficiency and rapidity of the D-DREAM algorithm through simulations and comparisons with existing MCMC algorithms. We then apply it to seven financial time series to highlight the presence of structural breaks. Under the assumption of Normal innovations, all seven studied series exhibit structural breaks and six entail recurrent regimes. With Student-$t$ innovations, we still find breaks, but less.

A forecasting exercise on the seven financial time series is carried out in order to empirically assess the forecast ability of CP-GJR-GARCH models with and without recurrent regimes. Our results show that these models can produce better forecasts than the fixed parameter GJR-GARCH model in about 25% of cases. We conjecture that a better performance may be obtained by reducing the number of parameters of the CP-GARCH model. Indeed, standard CP specifications assume that all GARCH parameters change when a structural break occurs. However it is likely that not all the parameters should change. Allowing for recurrent states is one way to decrease the parameter set but further research will be devoted to limit it even more, and the impact on predictions will be investigated.

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| Horizon $h$ | 1 | 2 | 5 | 10 | 20 | 40 | 50 |
|---|---|---|---|---|---|---|---|
| | | | S&P 500 | | | | |
| Normal | -1.33 | -2.65 | -7.18 | -14.22 | -28.92 | -59.99 | -75.68 |
| | (0.79) | (1.16) | (2.66) | (5.53) | (11.56) | (25.02) | (30.91) |
| Normal with rec | -1.33 | -2.68 | -7.25 | -14.38 | -29.25 | -60.55 | -76.36 |
| | (0.76) | (1.13) | (2.62) | (5.38) | (11.21) | (24.28) | (30.04) |
| Normal without rec | -1.33 | -2.68 | -7.25 | -14.38 | -29.25 | -60.57 | -76.40 |
| | (0.76) | (1.13) | (2.62) | (5.38) | (11.21) | (24.27) | (30.05) |
| Student | -1.30 | **-2.61** | **-7.11** | **-14.07** | **-28.66** | **-59.51** | **-75.22** |
| | (0.82) | (1.18) | (2.72) | (5.58) | (11.39) | (24.27) | (30.05) |
| Student with rec | -1.30 | -2.62 | -7.14 | -14.13 | -28.77 | -59.82 | -75.84 |
| | (0.87) | (1.21) | (2.81) | (5.64) | (11.37) | (24.31) | (30.10) |
| Student without rec | **-1.29** | -2.64 | -7.16 | -14.17 | -28.86 | -59.67 | -75.39 |
| | (0.78) | (1.14) | (2.69) | (5.48) | (11.19) | (23.83) | (29.57) |
| | | | DJIA | | | | |
| Normal | -1.32 | -2.69 | -6.99 | -13.59 | -27.85 | -58.07 | -73.35 |
| | (0.96) | (1.24) | (2.66) | (5.20) | (10.38) | (23.79) | (29.51) |
| Normal with rec | -1.34 | -2.69 | -7.07 | -13.74 | -28.04 | -58.69 | -74.38 |
| | (0.96) | (1.22) | (2.70) | (5.10) | (10.14) | (23.36) | (29.10) |
| Normal without rec | -1.34 | -2.68 | -7.06 | -13.74 | -28.04 | -58.72 | -74.40 |
| | (0.96) | (1.22) | (2.70) | (5.10) | (10.14) | (23.33) | (29.06) |
| Student | **-1.27** | **-2.65** | **-6.92** | **-13.44** | **-27.64** | **-57.54** | **-72.77** |
| | (0.86) | (1.23) | (2.75) | (5.33) | (10.51) | (23.67) | (29.29) |
| Student with rec | -1.32 | -2.65 | -7.01 | -13.61 | -27.79 | -58.09 | -73.69 |
| | (0.96) | (1.25) | (2.81) | (5.29) | (10.42) | (23.50) | (29.19) |
| Student without rec | -1.30 | -2.66 | -6.99 | -13.58 | -27.79 | -57.77 | -73.05 |
| | (0.90) | (1.22) | (2.78) | (5.23) | (10.34) | (23.17) | (28.79) |
| | | | NYSE | | | | |
| Normal | -1.33 | -2.72 | -7.15 | -14.00 | -28.50 | -59.96 | -75.90 |
| | (0.84) | (1.18) | (2.85) | (5.79) | (12.11) | (26.78) | (33.54) |
| Normal with rec | -1.34 | -2.73 | -7.23 | -14.16 | -28.78 | -60.17 | -76.22 |
| | (0.77) | (1.10) | (2.82) | (5.53) | (11.41) | (25.06) | (31.66) |
| Normal without rec | -1.34 | -2.73 | -7.23 | -14.16 | -28.77 | -60.16 | -76.21 |
| | (0.78) | (1.11) | (2.82) | (5.53) | (11.41) | (25.06) | (31.67) |
| Student | **-1.31** | **-2.70** | **-7.12** | **-13.87** | **-28.27** | **-59.55** | **-75.60** |
| | (0.79) | (1.20) | (2.92) | (5.78) | (11.77) | (25.35) | (31.70) |
| Student with rec | -1.34 | -2.72 | -7.22 | -14.06 | -28.51 | -60.00 | -76.16 |
| | (0.80) | (1.17) | (2.87) | (5.58) | (11.41) | (24.91) | (31.45) |
| Student without rec | -1.33 | -2.71 | -7.20 | -14.04 | -28.53 | -59.71 | -75.71 |
| | (0.77) | (1.11) | (2.86) | (5.58) | (11.38) | (24.73) | (31.23) |
| | | | Gold | | | | |
| Normal | **-1.43** | **-2.93** | -7.26 | -14.16 | -28.92 | -58.43 | -73.78 |
| | (0.72) | (1.14) | (2.75) | (4.57) | (7.66) | (13.74) | (17.28) |
| Normal with rec | -1.44 | -2.96 | **-7.24** | -14.21 | -28.96 | -58.63 | -74.09 |
| | (0.72) | (1.11) | (2.44) | (4.25) | (7.52) | (13.94) | (17.54) |
| Normal without rec | -1.48 | -3.00 | -7.29 | -14.20 | -28.96 | -58.65 | -74.10 |
| | (0.73) | (1.23) | (2.50) | (4.30) | (7.66) | (14.00) | (17.50) |
| Student | -1.48 | -3.03 | -7.34 | -14.00 | -28.72 | -58.33 | -73.67 |
| | (0.94) | (1.40) | (2.84) | (4.84) | (7.86) | (14.35) | (17.90) |
| Student with rec | -1.47 | -3.03 | -7.32 | **-13.97** | **-28.64** | **-58.25** | **-73.58** |
| | (0.90) | (1.36) | (2.81) | (4.76) | (7.84) | (14.25) | (17.74) |
| Student without rec | -1.47 | -3.03 | -7.33 | -13.98 | -28.67 | -58.33 | -73.67 |
| | (0.90) | (1.36) | (2.81) | (4.77) | (7.85) | (14.29) | (17.80) |

Table 17: Log-PMLs. The table shows the averages, over 49 forecasts, of the log-PMLs of the six classes of CP-GJR-GARCH(1,1) models at different forecast horizons. Standard deviations are reported between parentheses. The bold values indicate the highest average log-PMLs for each series and forecast horizon; "with rec" means that recurrent regimes are allowed, "without rec" means they are not.



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| Horizon $h$ | 1 | 2 | 5 | 10 | 20 | 40 | 50 |
|---|---|---|---|---|---|---|---|
| | | | Metals | | | | |
| Normal | -1.77 | -3.59 | -8.83 | -18.29 | -36.61 | -75.01 | -94.18 |
| | (1.17) | (1.56) | (2.73) | (5.79) | (10.68) | (18.80) | (23.33) |
| Normal with rec | -1.81 | -3.65 | -8.94 | -18.33 | -36.59 | -75.00 | -94.01 |
| | (1.15) | (1.53) | (2.76) | (5.54) | (10.39) | (17.78) | (21.59) |
| Normal without rec | -1.81 | -3.65 | -8.94 | -18.33 | -36.59 | -75.00 | -94.01 |
| | (1.15) | (1.53) | (2.76) | (5.55) | (10.39) | (17.78) | (21.59) |
| Student | **-1.71** | **-3.52** | **-8.76** | **-18.15** | -36.36 | **-74.22** | **-93.46** |
| | (1.02) | (1.47) | (2.72) | (5.49) | (10.17) | (17.65) | (22.01) |
| Student with rec | -1.75 | -3.57 | -8.82 | -18.18 | **-36.36** | -74.45 | -93.61 |
| | (1.03) | (1.46) | (2.72) | (5.41) | (10.13) | (17.75) | (22.16) |
| Student without rec | -1.75 | -3.57 | -8.82 | -18.18 | -36.36 | -74.45 | -93.61 |
| | (1.03) | (1.46) | (2.72) | (5.41) | (10.13) | (17.75) | (22.16) |
| | | | Boeing | | | | |
| Normal | -2.05 | -3.96 | -9.86 | -19.40 | -40.63 | -81.93 | -102.12 |
| | (0.88) | (1.12) | (2.37) | (4.07) | (10.38) | (18.68) | (22.24) |
| Normal with rec | -2.08 | -3.97 | -9.91 | -19.53 | -40.52 | -82.16 | -102.37 |
| | (0.92) | (1.17) | (2.44) | (4.21) | (9.43) | (19.05) | (22.65) |
| Normal without rec | -2.08 | **-3.95** | -9.91 | -19.49 | -40.37 | -81.79 | -101.96 |
| | (0.95) | (1.20) | (2.66) | (4.35) | (8.96) | (17.35) | (20.75) |
| Student | -2.05 | -3.97 | -9.84 | -19.31 | -40.00 | -80.90 | -100.97 |
| | (0.92) | (1.22) | (2.44) | (4.31) | (8.95) | (17.17) | (20.67) |
| Student with rec | **-2.05** | -3.96 | **-9.83** | **-19.31** | -40.07 | -80.94 | -100.97 |
| | (0.92) | (1.22) | (2.45) | (4.33) | (9.12) | (17.55) | (21.07) |
| Student without rec | -2.06 | -3.97 | -9.83 | -19.31 | **-39.96** | **-80.71** | **-100.72** |
| | (0.92) | (1.22) | (2.41) | (4.27) | (8.89) | (17.03) | (20.38) |
| | | | JP Morgan | | | | |
| Normal | -1.95 | -4.07 | -10.21 | -20.48 | -41.04 | -83.65 | -104.90 |
| | (0.80) | (1.51) | (4.27) | (6.97) | (12.35) | (25.47) | (32.11) |
| Normal with rec | -1.97 | -4.09 | -10.19 | -20.41 | -41.00 | -83.30 | -104.42 |
| | (0.86) | (1.55) | (4.32) | (7.08) | (12.60) | (26.36) | (33.32) |
| Normal without rec | -1.97 | -4.10 | -10.20 | -20.42 | -41.04 | -83.36 | -104.48 |
| | (0.86) | (1.54) | (4.32) | (7.08) | (12.60) | (26.36) | (33.31) |
| Student | **-1.92** | **-4.04** | **-10.05** | **-20.27** | **-40.72** | **-82.76** | **-103.83** |
| | (0.87) | (1.53) | (3.99) | (6.82) | (12.36) | (25.21) | (31.76) |
| Student with rec | -1.92 | -4.04 | -10.05 | -20.27 | -40.72 | -82.76 | -103.83 |
| | (0.87) | (1.53) | (3.99) | (6.82) | (12.36) | (25.21) | (31.76) |
| Student without rec | -1.92 | -4.04 | -10.05 | -20.27 | -40.72 | -82.76 | -103.83 |
| | (0.87) | (1.53) | (3.99) | (6.82) | (12.36) | (25.21) | (31.76) |

Table 18: Log-PMLs. The table shows the averages, over 49 forecasts, of the log-PMLs of the six classes of CP-GJR-GARCH(1,1) models at different forecast horizons. Standard deviations are reported between parentheses. The bold values indicate the highest average log-PMLs for each series and forecast horizon; "with rec" means that recurrent regimes are allowed, "without rec" means they are not.

## Appendices

### *A. Proof of Proposition 1*

In order to prove proposition 1, we need to show the convergence of the two conditional transition kernels for $\Theta$ and $\Gamma$ and then resort to the product of kernels principle (see Chib and Greenberg, 1995b). The proof of convergence in Vrugt et al. (2009) only requires minor adjustments to the present case to prove the convergence for the $\Theta$-block given $\Gamma$.



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We give here the proof for the Γ-block given Θ only.[7] It consists of four steps and is based on the fact that the chains do not converge separately but rather converge together towards a stationary distribution.

- Step 1. Recalling conditional detailed balance.

  In this first step, we recall the conditional detailed balance condition required for the existence of a stationary distribution in a N-component Metropolis algorithm. In order to have the stationary distribution as target distribution one needs to show that if $(\Gamma_1^j, ..., \Gamma_N^j) \sim \tilde{\pi}(.)$, with $\tilde{\pi}(.)$ the joint pdf of $(\Gamma_1, ..., \Gamma_N)$ at iteration $j$, then $(\Gamma_1^{j+1}, ..., \Gamma_N^{j+1}) \sim \tilde{\pi}(.)$ at iteration $j+1$. That is, one needs the conditional detailed balance condition to be satisfied for each chain $i$ such that, $\forall i$ (omitting $\Theta_i$ and $Y_T$ from the conditioning variables):

$$\begin{aligned}&\tilde{\pi}(\Gamma_i^j|\Gamma_1^{j+1}, ..., \Gamma_{i-1}^{j+1}, \Gamma_{i+1}^j, ..., \Gamma_N^j) \times K_i(\Gamma_i^{j+1}|\Gamma_1^{j+1}, ..., \Gamma_{i-1}^{j+1}, \Gamma_i^j, \Gamma_{i+1}^j, ..., \Gamma_N^j)\\ &= \tilde{\pi}(\Gamma_i^{j+1}|\Gamma_1^{j+1}, ..., \Gamma_{i-1}^{j+1}, \Gamma_{i+1}^j, ..., \Gamma_N^j) \times K_i(\Gamma_i^j|\Gamma_1^{j+1}, ..., \Gamma_{i-1}^{j+1}, \Gamma_i^{j+1}, \Gamma_{i+1}^j, ..., \Gamma_N^j),\end{aligned}$$

  with $K_i(.|.)$ the jumping kernel.

- Step 2. Proving the conditional detailed balance condition.

  In the second step, we prove the conditional detailed balance of the D-DREAM algorithm. The D-DREAM algorithm satisfies the detailed balance condition for the $\Gamma_i$-block *with respect to* $\pi(.)$ if

$$\begin{aligned}&\pi(\Gamma_i^j) \times K_i(\Gamma_i^{j+1}|\Gamma_1^{j+1}, ..., \Gamma_{i-1}^{j+1}, \Gamma_i^j, \Gamma_{i+1}^j, ..., \Gamma_N^j)\\ &= \pi(\Gamma_i^{j+1}) \times K_i(\Gamma_i^j|\Gamma_1^{j+1}, ..., \Gamma_{i-1}^{j+1}, \Gamma_i^{j+1}, \Gamma_{i+1}^j, ..., \Gamma_N^j).\end{aligned}$$

  Recall that the proposal $Z_i$ for an update of chain $i$ proceeds by random selection of pairs of other chains. Hence, the kernel of this update is a mixture of kernels which maintains conditional detailed balance with respect to $\pi(.)$ if each of its components does. Since $\xi$ follows a symmetric distribution and, for all $g$ and $h \in \{1, 2, ..., \delta\}$, the pairs $\{\Gamma_{r_1(g)}, \Gamma_{r_2(h)}\}$ and $\{\Gamma_{r_1(h)}, \Gamma_{r_2(g)}\}$ are equally likely, this condition is fulfilled by accepting the proposal with probability $\min\{\frac{f(Y_T|Z_i)p(Z_i)}{f(Y_T|\Gamma)p(\Gamma)}, 1\}$.[8] This also holds true with the *round*[.] operator since it does not alter the symmetry of the proposal distribution.[9]

- Step 3. Showing that the joint is the product of the marginals.

[7] A less detailed proof can also be found in Vrugt and Ter Braak (2011).

[8] As discussed above, the two possible issues in the proposal - that a break date falls outside the sample length [1,T-1] or that two break dates are inverted - are discarded by our prior distribution choice since the likelihood of the proposal $Z_i$ would then be equal to zero.

[9] Note that the *round[.]* operator must also insure symmetry for the particular case where one of the proposed date has a fractional part equal to .5, although it will virtually never happen in practice. The proposed parameter would then be rounded with probability .5 to the upper integer and .5 to the smaller integer.



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In the third step, we show that the joint distribution of the break date vectors of the $N$ chains is the product of the marginal distributions of the vector of break dates of each chain. As shown in step 2, the kernel $K_i(.|.)$ for updating chain $i$ satisfies the detailed balance condition of step 1 *with*, $\forall i = 1, ..., N$, $\tilde{\pi}(\Gamma_i^j|\Gamma_1^{j+1}, ..., \Gamma_{i-1}^{j+1}, \Gamma_{i+1}^j, ..., \Gamma_N^j) = \pi(\Gamma_i^j)$. Since the identity in step 1 is satisfied with the marginal distribution $\pi(\Gamma_i^j)$, the D-DREAM algorithm is an N-component Metropolis algorithm with joint stationary distribution $\tilde{\pi}(\Gamma_1, ..., \Gamma_N)$, such that

$$\tilde{\pi}(\Gamma_1, ..., \Gamma_N) = \pi(\Gamma_1) \times ... \times \pi(\Gamma_N).$$

- Step 4. Proving ergodicity and uniqueness.

  In the fourth step, we prove the ergodicity of the chains and convergence towards a unique stationary distribution. In order to prove convergence towards a unique stationary distribution, it is sufficient to show that the chains are irreducible and aperiodic. Irreducibility is guaranteed by the unbounded support of the distribution of $\xi$ in proposal (2) that necessarily covers the entire support [1,T-1], and aperiodicity is ensured by the random walk component generated by the same $\xi$. For ergodicity, it remains to show that the chains have a finite expected return time which is trivially observed given the discreteness of $\Gamma$ and the random terms in proposal (2).

In order to complete the proof one needs to combine the proof for the $\Gamma$-block above with the proof for the $\Theta$-block available in Vrugt et al. (2009) and recall the product of kernels principle. This principle states that the product of the transition kernels has an invariant density. The reader is referred to Chib and Greenberg (1995b) for further explanations.

*B. Proposal Distribution for the Computation of the Marginal Likelihood via Bridge Sampling*

We assume independence between the proposal distributions of $\Theta$ and $\Gamma$. Following Ardia et al. (2009) we work with mixtures of distributions. The proposal $q(\Theta)$ consists of two mixtures of Normal distributions. The expectations, the variance-covariance matrices and the weights are determined by the K-harmonic means algorithm based on the posterior draws but the variance-covariance matrices are multiplied by four in order to cover the entire posterior distribution. The proposal $q(\Gamma)$ is driven by a Poisson distribution defined as follows:

$$\begin{aligned} q(\Gamma) &= \prod_{i=1}^{K} q(\tau_i) \\ q(\tau_i) &= \bar{\tau}_i - \tilde{\lambda} + Po(\tilde{\lambda}), \end{aligned}$$

where $\bar{\tau}_i$ refers to the rounded posterior means of the distributions $\tau_i|Y_T$ and $\tilde{\lambda}$ is equal to four times the standard deviations of the same distributions. The proposal distribution implies an expectation equal to $\bar{\tau}_i$ and a variance of $\tilde{\lambda}$. Recall that the Jacobian of the transformation from break dates to durations is equal to one.



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*C. Posterior Results for the Best CP-GJR-GARCH(1,1) Model with Normal Distribution for the "Other Financial Series"*

| Parameters | Regime 1 | Regime 3 | Regime 2 | Regime 4 |
|---|---|---|---|---|
| $\mu$ | 0.046 | | 0.015 | |
| | (0.013) | | (0.020) | |
| $\omega$ | 0.054 | | 0.036 | |
| | (0.015) | | (0.006) | |
| $\alpha$ | -0.007 | | -0.009 | |
| | (0.008) | | (0.007) | |
| $\gamma$ | 0.105 | | 0.164 | |
| | (0.020) | | (0.015) | |
| $\beta$ | 0.829 | | 0.906 | |
| | (0.040) | | (0.010) | |
| Break Dates | 15-Nov-1996 | 07-Jun-2007 | 08-Aug-2003 | — |
| | (29.93) | (21.12) | (34.76) | — |

Table 1: Posterior Results for the Best CP-GJR-GARCH(1,1) Model on DJIA. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.

| Parameters | Regime 1 | Regime 2 | Regime 4 | Regime 3 | Regime 5 |
|---|---|---|---|---|---|
| $\mu$ | 0.059 | 0.047 | | -0.002 | |
| | (0.060) | (0.012) | | (0.018) | |
| $\omega$ | 0.357 | 0.029 | | 0.044 | |
| | (0.093) | (0.006) | | (0.006) | |
| $\alpha$ | 0.025 | -0.019 | | -0.014 | |
| | (0.080) | (0.010) | | (0.007) | |
| $\gamma$ | 0.185 | 0.148 | | 0.178 | |
| | (0.159) | (0.024) | | (0.017) | |
| $\beta$ | 0.297 | 0.862 | | 0.897 | |
| | (0.137) | (0.021) | | (0.011) | |
| Break Dates | 28-Jan-1992 | 26-Dec-1996 | 17-Jan-2007 | 08-Sep-2003 | — |
| | (29.215) | (24.198) | (18.73) | (64.838) | — |

Table 2: Posterior Results for the Best CP-GJR-GARCH(1,1) Model on NYSE. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.



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| Parameters | Regime 1 | Regime 4 | Regime 2 | Regime 3 |
|---|---|---|---|---|
| $\mu$ | 0.024 | | -0.039 | 0.108 |
| | (0.009) | | (0.018) | (0.093) |
| $\omega$ | 0.001 | | 0.088 | 0.277 |
| | (0.000) | | (0.018) | (0.096) |
| $\alpha$ | 0.073 | | 0.206 | 0.593 |
| | (0.006) | | (0.024) | (0.215) |
| $\gamma$ | -0.059 | | -0.159 | 0.730 |
| | (0.006) | | (0.023) | (0.198) |
| $\beta$ | 0.958 | | 0.721 | 0.142 |
| | (0.003) | | (0.042) | (0.074) |
| Break Dates | 22-Nov-1996 | — | 27-Aug-2001 | 14-Nov-2001 |
| | (2.59) | — | (13.15) | (12.66) |

Table 3: Posterior Results for the Best CP-GJR-GARCH(1,1) Model on Gold. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.

| Parameters | Regime 1 | Regime 2 | Regime 3 | Regime 4 |
|---|---|---|---|---|
| $\mu$ | -0.048 | 0.002 | 0.159 | 0.004 |
| | (0.026) | (0.016) | (0.038) | (0.049) |
| $\omega$ | 0.277 | 0.157 | 0.010 | 0.364 |
| | (0.043) | (0.040) | (0.007) | (0.125) |
| $\alpha$ | 0.252 | 0.061 | 0.103 | 0.053 |
| | (0.062) | (0.015) | (0.015) | (0.021) |
| $\gamma$ | -0.141 | 0.046 | -0.110 | 0.058 |
| | (0.043) | (0.020) | (0.015) | (0.030) |
| $\beta$ | 0.241 | 0.733 | 0.960 | 0.827 |
| | (0.095) | (0.059) | (0.010) | (0.044) |
| Break Dates | 19-Jul-1993 | 27-May-2003 | 07-Nov-2006 | — |
| | (13.51) | (84.23) | (27.33) | — |

Table 4: Posterior Results for the Best CP-GJR-GARCH(1,1) Model on Metals. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.



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| Parameters | Regime 1 | Regime 4 | Regime 2 | Regime 3 |
|---|---|---|---|---|
| $\mu$ | 0.050 | | -0.099 | 0.052 |
| | (0.022) | | (0.218) | (0.071) |
| $\omega$ | 0.041 | | 1.551 | 1.458 |
| | (0.008) | | (0.501) | (0.395) |
| $\alpha$ | 0.013 | | -0.127 | 0.129 |
| | (0.005) | | (0.019) | (0.034) |
| $\gamma$ | 0.054 | | 0.104 | 0.064 |
| | (0.009) | | (0.017) | (0.046) |
| $\beta$ | 0.947 | | 0.928 | 0.609 |
| | (0.007) | | (0.052) | (0.077) |
| Break Dates | 04-Jun-1998 | — | 01-Dec-1998 | 29-Aug-2002 |
| | (4.83) | — | (14.48) | (82.92) |

Table 5: Posterior Results for the Best CP-GJR-GARCH(1,1) Model on Boeing. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.

| Parameters | Regime 1 | Regime 3 | Regime 2 |
|---|---|---|---|
| $\mu$ | 0.031 | | 0.056 |
| | (0.027) | | (0.029) |
| $\omega$ | 0.069 | | 0.089 |
| | (0.011) | | (0.044) |
| $\alpha$ | 0.013 | | 0.020 |
| | (0.005) | | (0.013) |
| $\gamma$ | 0.097 | | 0.084 |
| | (0.010) | | (0.038) |
| $\beta$ | 0.930 | | 0.854 |
| | (0.006) | | (0.056) |
| Break Dates | 08-Sep-2003 | — | 13-Apr-2007 |
| | (32.48) | — | (37.29) |

Table 6: Posterior Results for the Best CP-GJR-GARCH(1,1) Model on JP Morgan. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.



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*D. Posterior Results for the Best CP-GJR-GARCH(1,1) Model with Student-t distribution for the "Other Financial Series"*

| Parameters | Regime 1 |
|---|---|
| $\mu$ | 0.047 |
| | (0.010) |
| $\omega$ | 0.008 |
| | (0.002) |
| $\alpha$ | 0.004 |
| | (0.005) |
| $\gamma$ | 0.085 |
| | (0.009) |
| $\beta$ | 0.925 |
| | (0.008) |
| *df* | 6.999 |
| | (0.642) |

Table 7: Posterior Results for the Best CP-GJR-GARCH(1,1) Model with Student-$t$ Distribution on DJIA. The table displays the posterior means and standard deviations (between parentheses) of the model parameters.

| Parameters | Regime 1 | Regime 2 |
|---|---|---|
| $\mu$ | 0.050 | 0.043 |
| | (0.012) | (0.012) |
| $\omega$ | 0.025 | 0.013 |
| | (0.010) | (0.002) |
| $\alpha$ | 0.004 | -0.013 |
| | (0.009) | (0.005) |
| $\gamma$ | 0.067 | 0.116 |
| | (0.021) | (0.011) |
| $\beta$ | 0.807 | 0.925 |
| | (0.061) | (0.008) |
| *df* | 4.856 | 9.154 |
| | (0.633) | (1.182) |
| Break Dates | 10-Sept-96 | — |
| | (44.70) | — |

Table 8: Posterior Results for the Best CP-GJR-GARCH(1,1) Model with Student-$t$ Distribution on NYSE. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.



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| Parameters | Regime 1 | Regime 2 |
|---|---|---|
| $\mu$ | -0.010 | 0.100 |
| | (0.008) | (0.018) |
| $\omega$ | 0.003 | 0.001 |
| | (0.001) | (0.002) |
| $\alpha$ | 0.043 | 0.049 |
| | (0.008) | (0.006) |
| $\gamma$ | -0.013 | -0.036 |
| | (0.008) | (0.007) |
| $\beta$ | 0.908 | 0.957 |
| | (0.014) | (0.005) |
| *df* | 3.146 | 5.689 |
| | (0.215) | (0.681) |
| Break Dates | 7-Feb-02 | — |
| | (43.11) | — |

Table 9: Posterior Results for the Best CP-GJR-GARCH(1,1) Model with Student-$t$ Distribution on Gold. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.

| Parameters | Regime 1 | Regime 2 | Regime 3 |
|---|---|---|---|
| $\mu$ | -0.052 | 0.005 | 0.110 |
| | (0.022) | (0.015) | (0.029) |
| $\omega$ | 0.129 | 0.063 | 0.044 |
| | (0.031) | (0.016) | (0.017) |
| $\alpha$ | 0.181 | 0.039 | 0.057 |
| | (0.058) | (0.009) | (0.011) |
| $\gamma$ | -0.099 | 0.022 | -0.010 |
| | (0.066) | (0.013) | (0.012) |
| $\beta$ | 0.326 | 0.844 | 0.916 |
| | (0.116) | (0.030) | (0.016) |
| *df* | 4.590 | 9.914 | 8.252 |
| | (0.877) | (1.488) | (1.254) |
| Break Dates | 9-Jul-93 | 15-Aug-03 | — |
| | (40.56) | (70.42) | — |

Table 10: Posterior Results for the Best CP-GJR-GARCH(1,1) Model with Student-$t$ Distribution on Metals. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.



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| Parameters | Regime 1 | Regime 3 | Regime 2 |
|---|---|---|---|
| $\mu$ | 0.043 | | 0.013 |
| | (0.021) | | (0.058) |
| $\omega$ | 0.027 | | 1.149 |
| | (0.006) | | (0.331) |
| $\alpha$ | 0.018 | | 0.071 |
| | (0.004) | | (0.026) |
| $\gamma$ | 0.038 | | 0.032 |
| | (0.007) | | (0.035) |
| $\beta$ | 0.940 | | 0.526 |
| | (0.007) | | (0.133) |
| $df$ | 7.653 | | 4.886 |
| | (0.778) | | (0.684) |
| Break Dates | 1-May-97 | — | 7-Feb-02 |
| | (68.65) | — | (244.35) |

Table 11: Posterior Results for the Best CP-GJR-GARCH(1,1) Model with Student-$t$ Distribution on Boeing. The table displays the posterior means and standard deviations (between parentheses) of the model parameters and break dates.

| Parameters | Regime 1 |
|---|---|
| $\mu$ | 0.038 |
| | (0.020) |
| $\omega$ | 0.017 |
| | (0.004) |
| $\alpha$ | 0.021 |
| | (0.005) |
| $\gamma$ | 0.063 |
| | (0.008) |
| $\beta$ | 0.929 |
| | (0.008) |
| $df$ | 7.503 |
| | (0.690) |

Table 12: Posterior Results for the Best CP-GJR-GARCH(1,1) Model with Student-$t$ Distribution on JP Morgan. The table displays the posterior means and standard deviations (between parentheses) of the model parameters.



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